# COUNTING THE ACTIVE DEGREES OF FREEDOM OF A TRAINING RUN: WHAT A SINGLE SCALAR LOG CAN AND CANNOT MEASURE

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## ABSTRACT

Over a window of training, a neural network's parameter trajectory moves on a set far smaller than the parameter space. We call the dimension of that set the active dimension of the window: a count of independent components, distinct from the number of directions the optimiser may move in, from the rank of the map to the outputs, and from the effective rank of the trajectory covariance. A stored trajectory measures it directly but is expensive; a scalar log is free, and we ask what survives the projection onto one. The instrument is standard, delay reconstruction followed by a maximum likelihood intrinsic dimension, and its hazard is that it returns a plausible number when no invariant set exists at all. On six constructed systems, ending with a linear head on a frozen image classifier, recovery on withheld ranks and seeds reaches a mean absolute error of $0.87$ components up to eight directions, and two diagnostics computed from the series alone flag the two regimes in which the value may not be read as a count. Applied to grokking they place both standard experimental settings outside the admissible regime, for two different reasons which they identify correctly. Paying instead for the trajectory, the effective rank collapses by a median factor of seven near generalisation in four regularised runs and then re-expands, which is the simplification the mechanistic accounts of grokking lead one to expect. Their controls fall too, so what marks the transition is not the depth of the fall but that it is sharp, locked to the transition, and reversed.

## 1 INTRODUCTION

A neural network is trained in a parameter space of dimension $P$, but the trajectory the optimiser follows does not fill it. Over a window it moves on a much smaller set, and the size of that set changes over a run. We call its dimension the *active dimension* of the window and define it in equation (1). It is a count of independent components: if the optimiser is carried by $r$ independent quasiperiodic phases the orbit closes onto an $r$-torus and the active dimension is $r$, with no tunable scale. It is neither the dimension of the subspace the optimiser is confined to, which the parameterisation fixes (Li et al., 2018), nor the rank of the map from it to the outputs, nor the effective rank of the trajectory covariance, which anisotropy depresses below the count; section 3.1 separates the four.

Knowing it would settle questions that are currently argued with whichever of those four quantities is cheapest to compute: whether an apparent simplification of training is a real reduction in degrees of freedom or an artefact of a shrinking scale, and whether a phase transition can be seen without instrumenting the run. Measuring it directly means storing the trajectory, which is expensive; a scalar log costs nothing, and a run already keeps several. We therefore ask how much of the active dimension survives the projection onto one such series. The instrument is standard, delay reconstruction (Takens, 1981; Sauer et al., 1991) followed by the pooled Levina–Bickel estimator (Levina & Bickel, 2004; MacKay & Ghahramani, 2005). Its hazard is that it returns a plausible number when no invariant set exists at all: power-law noise yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989), and a decaying transient, which is what a converging run looks like, yields a large one. Most of the work below therefore goes into deciding which case one has. Section 4 fixes a validation protocol before any result, and two of our six systems fail one of its controls by reproducing their reported result with no training taking place.

**Contributions.**

1. We separate four quantities that are frequently identified with one another: the dimension of the subspace the optimiser is confined to, the rank of the map from it to the outputs, the number of independent components of the set it recurrently visits, and the effective rank of its covariance (section 3.1). Only the third is estimated here, and the fourth is measured beside it on every system so that the two cannot be conflated.
2. We evaluate the estimator on six systems of increasing realism, ending with a constrained head on a network trained on image data (section 5). It recovers the active dimension to a mean absolute error of $0.87$ components up to eight directions, without per-system calibration, on withheld ranks and seeds, and locates the ceiling above which it cannot (section 5.2).
3. We measure the conditions the absolute value requires (section 6), including a control that separates the effective rank from the manifold dimension, and validate two diagnostics that flag the two failure regimes from the series alone.
4. Applied to grokking (section 7), the diagnostics place both published settings outside the regime in which a scalar log could be read as a count, while the trajectory effective rank, measured directly, collapses near generalisation and re-expands in four regularised runs.

## 2 Related work

**Reconstruction and intrinsic dimension.** Delay reconstruction rests on Takens' theorem and its extensions (Takens, 1981; Sauer et al., 1991; Stark, 1999; Stark et al., 2003), is used on empirical series (Sugihara & May, 1990; Sugihara et al., 2012), and takes its lag and dimension from mutual information, false nearest neighbours or singular spectra (Fraser & Swinney, 1986; Kennel et al., 1992; Cao, 1997; Broomhead & King, 1986). We use pooled maximum likelihood (Levina & Bickel, 2004; MacKay & Ghahramani, 2005), not the alternatives (Grassberger & Procaccia, 1983; Facco et al., 2017; Camastra & Staiano, 2016); a finite record bounds them all (Eckmann & Ruelle, 1992). Oversampling (Theiler, 1986) and power-law noise (Osborne & Provenzale, 1989) produce spurious dimensions, and surrogate tests (Theiler et al., 1992; Schreiber & Schmitz, 1996) detect nonlinear structure but not its amount, missing the transient regime of section 3.3 entirely.

**Dimension in optimisation.** Random-subspace training (Li et al., 2018; Aghajanyan et al., 2021) reports the smallest $k$ that still solves a task, a property of the task rather than of a window; we borrow the parameterisation and measure something else inside it. The closest existing statement to the active dimension is that gradient descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018; Sagun et al., 2017). The intrinsic dimension of data or of representations (Ansuini et al., 2019; Pope et al., 2021) is a point cloud's, not a trajectory's, and the participation ratio we adopt (Gao et al., 2017) is not the spectral-entropy quantity that shares its name (Roy & Vetterli, 2007). The one prior measurement of our own object we are aware of is incidental: two principal components carry over $98\,\%$ of the variance of a grokking trajectory (Notsawo Jr. et al., 2023). All of these need the weights, the gradients or the activations, and the nearest precedent for a scalar computed during training and then acted on is the gradient noise scale (McCandlish et al., 2018). Mini-batch training near a minimum is modelled as a stochastic process (Mandt et al., 2017), the third regime of table 1; whether the oscillations of full-batch descent at the edge of stability (Cohen et al., 2021) recur in the sense the first regime needs is open.

**Grokking.** Delayed generalisation (Power et al., 2022) is studied in two settings and we use both. The first is regularised and, in our version, stochastic: the transformer configuration of Nanda et al. (2023), also used by Liu et al. (2023), on modular addition and on the $S_5$ task of Power et al. (2022). The second is the unregularised full-batch gradient descent of Gromov (2023), whose closed form supplies an analytic reference, on the modular polynomials and representability criterion of Doshi et al. (2024), which supply label-matched pairs; Appendix O states where each departs from its source. Explanations sit in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Liu et al., 2022; Barak et al., 2022) or in a lazy-to-rich transition (Kumar et al., 2024). Closest to us, Notsawo Jr. et al. (2023) predict grokking from early loss oscillations (Thilak et al., 2022) and Merrill et al. (2023) track competing subnetworks. We estimate a dynamical quantity with an independently measured ground truth.

# 3 PROBLEM STATEMENT

## 3.1 FOUR QUANTITIES, AND THE ONE WE ESTIMATE

Let $\boldsymbol{\theta}_t \in \mathbb{R}^P$ be the parameter vector at step $t$, confined by construction or by the dynamics to an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{V}^\top \boldsymbol{c}$ with $\boldsymbol{V} \in \mathbb{R}^{k\times P}$ orthonormal. Over a window $\mathcal{W}$ the *available* dimension is $k$, fixed by the parameterisation. The *functional* dimension, the rank of $\partial \boldsymbol{f}(\boldsymbol{c})/\partial \boldsymbol{c}$ for $\boldsymbol{f}$ the outputs on a fixed probe set, bounds how many of those directions can change the computed function: a bound we report and do not estimate. The *active* dimension is

$$d_{\text{act}}(\mathcal{W}) \;=\; \dim \mathcal{M}_{\mathcal{W}}, \tag{1}$$

where $\mathcal{M}_{\mathcal{W}}$ is the set of states the trajectory revisits inside the window: the support of the empirical measure of $\{\boldsymbol{c}_t : t \in \mathcal{W}\}$, coarsened to a scale $\varepsilon$, over a window long compared with the recurrence time. Both qualifications are needed. The closure of a finite orbit is a curve, so the torus is only the set an orbit has had time to fill; and a neighbour statistic sees that set at the scale of its own neighbourhoods, not at scale zero. If the parameters depend on $r$ independent quasiperiodic phases, $\boldsymbol{c}_t = F(\varphi_1(t), \ldots, \varphi_r(t))$, the set is an $r$-torus and $d_{\text{act}} = r$ for every $\varepsilon$ below the smallest mode amplitude. Above that the count falls: a property of the estimand, not a defect of the estimator, and section 6.4 measures it. The set must also exist, and in two of the three regimes of section 3.3 none does at any $\varepsilon$. That is why the diagnostics of section 3.4 come before any number.

A fourth quantity is measured throughout and is not the active dimension. For a non-negative spectrum $\lambda_1, \ldots, \lambda_n$ the participation ratio

$$\mathrm{PR}(\lambda) \;=\; \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \tag{2}$$

equals $m$ exactly when $m$ of the $\lambda_i$ are equal and the rest vanish. On $\operatorname{spec}\operatorname{Cov}\{\boldsymbol{c}_t : t \in \mathcal{W}\}$ it gives the *effective rank* $\mathrm{PR}^{\text{pos}}(\mathcal{W})$, which is what the term means below: $\mathrm{PR}^{\text{det}}$ is the same after the per-window linear detrend a drifting trajectory needs, and $\mathrm{PR}^{\text{upd}}$ on the increments. The effective rank is not a count: it equals $r$ when $r$ directions carry comparable variance and falls below $r$ when they do not, while $d_{\text{act}}$ stays at $r$ (section 6.4). It verifies that a construction excites the $r$ directions claimed, and section 7.3 measures it. All four differ measurably: in section 5.4 the available and functional dimensions are both 10, and removing the preconditioner leaves them and the injected rank intact while stochastic forcing takes the effective rank from 7.98 to 6.43.

## 3.2 OBSERVATION MODEL AND ESTIMATOR

We observe a scalar series $x_t = \phi(\boldsymbol{s}_t)$, with $\boldsymbol{s}_t$ the full optimiser state and $\phi$ fixed before the run, and reconstruct $\boldsymbol{y}_t = (x_t, \ldots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant manifold of integer dimension $d$ this map is an embedding once $E > 2d$ (Takens, 1981); the prevalence form replaces $d$ by the box-counting dimension, which need not be an integer, and adds conditions on periodic orbits (Sauer et al., 1991). Genericity is not verifiable for a $\phi$ fixed in advance, so every result below is reported over a family of observers. We estimate the intrinsic dimension of $\{\boldsymbol{y}_t\}$ by maximum likelihood on nearest-neighbour distances (Levina & Bickel, 2004). With $r_1^{(i)} \le \cdots \le r_m^{(i)}$ the distances from point $i$ to its $m$ nearest neighbours, subject to a Theiler exclusion $W_T$ on their separation in time (Theiler, 1986; Kantz & Schreiber, 2004), the local estimate is

$$\hat{d}_i \;=\; \left[ \frac{1}{m-1} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}} \right]^{-1}, \tag{3}$$

and the $N$ local estimates combine in two ways. Averaging them, as Levina & Bickel (2004) propose, gives $\hat{d}_{\text{LB}} = N^{-1} \sum_i \hat{d}_i$. Pooling the likelihood first, as MacKay & Ghahramani (2005) argue one should, averages the reciprocals,

$$\hat{d}_{\text{MG}}^{-1} \;=\; \frac{1}{N(m-1)} \sum_{i=1}^{N} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}. \tag{4}$$

We write LB and MG for the two, *the estimate* for what MG returns, *the estimator* for the procedure. Under the local Poisson model behind both, $\hat{d}_i^{-1}$ is unbiased where $\hat{d}_i$ is not, so LB should exceed

Table 1: What the orbit fills in each dynamical regime, whether the active dimension of equation (1) is defined there, and what the estimator returns on the systems of section 5. Only the first regime supports reading the estimate as a count.

| regime | what the orbit fills | $d_{\text{act}}$ | estimate here |
|---|---|---|---|
| deterministic, recurrent | an $r$-torus | $r$ | $r$ |
| deterministic, transient | a curve, never revisited | undefined | $\approx 29$, at every $r$ |
| stochastically driven | no low-dimensional invariant set | undefined | set by $E_{\max}$, not by $r$ |

MG by $(m-1)/(m-2)$, under six per cent at the $m = 20$ used here; table 7 reports both and Appendix A gives the algorithm as implemented. Three further statistics accompany every estimate, since each names an effect that can produce a dimension signal with no geometry: the *roughness* $\operatorname{std}(\Delta x)/\operatorname{std}(x)$, the *linear participation ratio* $\mathrm{PR}_{\text{delay}}$ of equation (2) on $\operatorname{Cov}\{\boldsymbol{y}_t\}$, and the *spectral participation ratio* of equation (2) on the power spectrum of $x$.

### 3.3 Dynamical regimes

Two conditions must hold and they are not the same. The theorems require a deterministic flow on a compact invariant set, which is what makes $\mathcal{M}_{\mathcal{W}}$ exist. The neighbour statistic needs more: the record must visit that set densely enough for near neighbours to be recurrences rather than temporal neighbours. The Theiler exclusion enforces that; section 3.4 tests it. An optimiser near a solution may be in any of the three regimes of table 1, and only the first meets the first condition.

The third regime is what mini-batch gradient noise produces, and theory serves it worse than it looks. The embedding theorems for forced systems (Stark, 1999; Stark et al., 2003) cover it in part: the update may depend on the state arbitrarily, and their iterated-function-system form admits a driver selecting among finitely many maps, as sampling a finite dataset does. But they deliver an embedding *fibre by fibre*, one per realisation of the drive, and a single run crosses fibres. Its cloud is not confined to the image of any one fibre, and no invariant measure exists there to sample. Coloured noise alone yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989). The second regime is less often noticed. A decaying transient traces a curve it never returns to, so it has no $\mathcal{M}_{\mathcal{W}}$; its effective rank measures 1.03 to 1.06 in section 5.4 only because the curve is nearly straight.

### 3.4 Admissibility diagnostics

Two statistics computed from the series alone flag the two failure regimes. The **identifiability ratio** $\rho_{\text{ident}} = \hat{d}_{\text{MG}}(2E_{\max})/\hat{d}_{\text{MG}}(E_{\max})$ is near unity when a dimension is resolvable and grows when the estimate is a property of the embedding space rather than of the data, which catches the stochastic regime. On a transient it is near unity too, for the wrong reason, and the **trend-crossing count** separates the two: the number of sign changes of the window's residual about its least-squares line, near zero there. It tests non-monotonicity of the series, not recurrence in the reconstructed space, so the pair must be read together. A third quantity is reported beside them but tests no regime. The **degeneracy flag** fires when more than one per cent of neighbour distances or per-point log-ratio sums reach their numerical floors: a conditioning guard against quantised or constant observers. Reference values are in section 6.4.

## 4 Validation protocol

Six requirements were fixed before any result, and each is named so that table 2 can say which a system fails. (1) *Constructed truth*: every system is built with $r$ independent quasiperiodic phases, so $d_{\text{act}} = r$, and the effective rank is then measured on the recorded trajectory to confirm all $r$ are comparably excited. (2) *Frozen configuration*: the estimator is selected once, on a grid of estimator parameters only, since a grid over system parameters changes the data when one is chosen. (3) *Disjoint split*: in seed and in rank, since withholding seeds alone leaves the excitation geometry shared whenever it is a function of $r$. (4) *Silence control*: a zero learning rate must silence the observer, which otherwise reads the drive through the residual rather than through the optimiser. (5) *Companion statistics*: the three of section 3.2 are reported on every call, and one performing comparably is reported as such. (6) *Matched bandwidth*: across ranks, or a construction in which each added direction adds a higher frequency lets the roughness alone order the rank.

# 5 EVALUATION ON SYSTEMS WITH A KNOWN ACTIVE DIMENSION

Six systems of increasing realism are used. In each the orbit closes onto an $r$-torus by construction, the measured effective rank confirms it, and the estimator sees one scalar series.

## 5.1 AN OSCILLATING DIAGONAL MATRIX

The first system has no optimiser and no feedback, so a negative result is decisive. In $\boldsymbol{W}(t) = \mathrm{diag}(b_i + \delta_i(t))$, $r$ coordinates oscillate at rationally independent frequencies over offsets that must be non-zero. On its two held-out seeds the mean absolute error over $r = 1, \ldots, 10$ is $0.30$ and $0.69$. It also supplies the geometry control: $r$ coordinates driven from one phase have one degree of freedom, and the estimate separates them from four independent oscillators by a factor of $3.4$, against $5\,\%$ for the roughness. Smoothness cannot do that work.

## 5.2 THE CEILING

The estimator saturates, and the ladder locates where. The matrix error rises to $1.51$ at ten directions; a $D = 64$ linear model and a logistic model order the rank to twenty but saturate above about twelve; and the image-data system at twenty directions saturates at $15.1$. The condition of section 3.2 is $E > 2d$, so $E_{\max} = 20$ cannot resolve more than ten components and $E_{\max} = 40$ cannot resolve twenty. The observed ceilings are consistent with that: above about nine at $E_{\max} = 20$, and $15.1$ against a measured $19.99$ at $E_{\max} = 40$, where $E = 2d$ exactly and the condition fails. The twenty-direction result is therefore not an anomaly; that design cannot recover twenty components. The explanation is not settled, since a finite record also bounds the measurable dimension (Eckmann & Ruelle, 1992), comparably so at the $8\,000$-sample window, and separating the two needs a sweep of $E_{\max}$ at fixed record length and the reverse, which we have not run.

## 5.3 THE SILENCE CONTROL

Requirement 4 asks that a zero learning rate silence the observer. Most published dimension-on-training-logs analyses do not run it; it invalidated two of our six systems. A trainable $\boldsymbol{z} \in \mathbb{R}^r$ optimised through a frozen nonlinear decoder, and a two-layer perceptron restricted to $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{U}\boldsymbol{z}$, read naively both succeed. But at zero learning rate, where no training occurs, each reproduces its reported result almost exactly, the two series correlating above $0.98$ at every rank, because the drive acts on the targets and so reaches the observers through the residual rather than through the optimiser. The same control rejects the instantaneous mini-batch loss of section 5.4, an observer that otherwise scores among the best three. The two regression systems above are excluded for a different reason: their configuration was selected on a grid that varied the system as well as the estimator, in breach of requirement 2, so table 2 locates their ceiling but measures no recovery.

## 5.4 A CONSTRAINED HEAD ON A NETWORK TRAINED ON IMAGE DATA

The sixth system uses real inputs: a tanh network trained on the scikit-learn handwritten digits and frozen, with a linear head confined to a $k$-dimensional affine subspace under softmax cross-entropy, so the curvature is data-dependent and all dynamics occur in $\boldsymbol{c}$. The excitation is made $r$-dimensional by modulating the loss weights of $r$ of twelve fixed data groups and orthogonalising the resulting gradient directions (Appendix F), after which the measured effective rank equals $r$ to within $0.013$ components. Nine excitation arms are run, spanning the three regimes of table 1; figure 1 shows five.

Twelve observers are recorded (Appendix B); the estimator's five parameters were frozen on four of them and on reserved seeds and ranks, so what follows is out-of-sample in rank and seed throughout and in observer for six of the ten scored. In the deterministic recurrent regime the mean absolute error is $0.87$ components run by run and $0.47$ after aggregating seeds, at a median slope of $0.92$ and a perfect rank correlation for all ten. The two parameter norms, which are what a run ordinarily logs, sit in the middle of that range and are also the only observers whose roughness orders the withheld

Table 2: The systems, ordered by increasing realism. The last column names the requirements of section 4 each row fails. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run's sliding windows, except the last row, which has one window per run; $\rho$ is over ranks after a median over seeds. The two oscillating-matrix figures are its two held-out seeds.

| system | truth | range | MAE | $\rho$ | protocol |
|---|---|---|---|---|---|
| oscillating matrix | constructed | 1–10 | 0.30 / 0.69 | $+0.99$–$+1.00$ | fails 2, 5 |
| online linear regression | constructed | 1–20 | 1.34 | $+0.98$ | fails 2 |
| logistic regression | constructed | 1–20 | 1.33 | $+0.99$ | fails 2 |
| frozen nonlinear decoder | constructed | 1–20 | 2.13 | $+0.92$ | fails 4 |
| perceptron in a $k$-subspace | constructed | 1–20 | 2.06 | $+0.94$ | fails 4 |
| image data, parameter subspace | measured | 1–8 | 0.87 | $+1.00$ | **complete** |
| image data, parameter subspace | measured | 1–20 | 1.62 | $+0.97$ | fails 3 |
| image data, function subspace | measured | 1–8 | 0.50 | $+0.99$ | fails 2, 3, 4 |

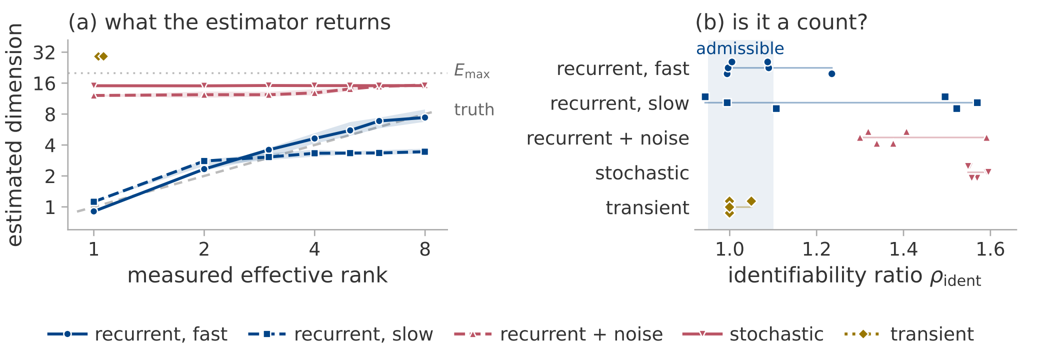


Figure 1: The evaluation of section 5.4: a frozen tanh network on the digits data, a linear head confined to ten directions of which $r$ are excited, five of the nine arms. Colour is the regime of table 1; "fast" and "slow" are the two recurrent drive rates and "+ noise" the fast drive with weak noise added. (a) The estimate against the measured effective rank, which the construction holds at $r$; the marker is the median of the forty values behind each point, four seeds by ten observers, and the band their interquartile range. The dashed diagonal is the truth and the dotted line $E_{\max}$, a reference and not a bound: no clamping is applied and the transient arm sits above it, its seven points piled at $x \approx 1.05$ because its measured rank is about one at every $r$. (b) The identifiability ratio, which separates the regimes with no ground truth; the shaded column is the admissible band. It is *not* computed on the runs of (a): the sweep populates it for seed 0, three observers and $r \in \{2, 6\}$ only, so each regime rests on the six values plotted. The transient is the one regime it passes, hence the trend-crossing count of section 3.4.

ranks, so for the cheapest observer a smoothness explanation cannot be excluded. Twenty available directions give the same picture with a lower ceiling, saturating at 15.1 against a measured 19.99 at Spearman $+0.97$; that arm is exploratory, having no seed split and a Theiler exclusion smaller than its own embedding span (Appendix N). A second construction in function space is the most accurate row of table 2, but its rank is set by an external teacher.

## 6 Conditions of validity

### 6.1 Dynamical regime

Figure 1 evaluates the estimator in each regime of table 1 at the same nominal rank, observer and matched octave. Under stochastic excitation the estimate is flat in the rank and the error exceeds ten components; it responds to $E_{\max}$, the correlation time and the record length. Weak noise destroys the absolute value on a recurrent trajectory and leaves the ordering: at a quarter of the pure noise amplitude, injected in the same $r$ directions and leaving the effective rank at $r$, the error rises tenfold while the Spearman correlation stays above $+0.96$. A decaying transient has no active dimension at all, returning near no state it has left, but it gives a reproducible estimate of about 29. Exceeding $E_{\max}$ is not a bound failing: no clamping is applied (Appendix A), and a maximum likelihood fit exceeds the embedding dimension when its model is violated.

### 6.2 THE DELAY LAG

The delay window spans $(E_{\max}-1)\tau$ samples and must cover a real fraction of the oscillation period, or the delay coordinates are collinear and the torus does not unfold. That the window rather than the lag matters is standard (Casdagli et al., 1991). On a fixed torus of period 400 at eight active directions, sweeping $\tau$ alone moves the estimate from 2.60 to 63.35, with 8.53 at the span of four tenths of a period that is right (Appendix E). The correct lag follows from the oscillation period, which a controlled experiment knows and a training log does not. We do not solve this. The lag applied in section 7 is the frozen $\tau = 4$ of Appendix C, transplanted from a calibration system of drive period 16; the autocorrelation time of the parameter norm on those logs is 161 to 858, so a matched lag would have been an order of magnitude larger. That alone forbids reading the absolute value there, whatever the dynamical regime. The identifiability ratio inherits this. Doubling $E_{\max}$ at fixed $\tau$ doubles the delay span too, so a large value means either that the estimate is a property of the embedding space rather than of the data or that the lag is mismatched. We cannot separate the two readings, and on the grokking logs both are live.

### 6.3 NUISANCE FACTORS

At a fixed active dimension of four we vary seven factors that leave it unchanged, over five seeds and twelve observers, and compare the shift against a real four-component change (Appendix E). Two matter. The smoothness control passes on eight of the ten observers: an octave shift of the drive band changes the roughness fourfold and moves the estimate only to the level of seed-to-seed variability. It fails on the two parameter norms, whose roughness orders the withheld ranks perfectly. A *ramped* observer gain, unlike a constant rescaling, moves the estimate by 1.16 components: a run whose parameter norm grows drifts by about one component with nothing dimensional occurring, the nuisance that matters most in section 7.

### 6.4 THE ESTIMAND IS A COUNT, NOT AN EFFECTIVE RANK

Every construction in section 5 equalises the drive amplitudes, making the active dimension and the effective rank of section 3.1 numerically equal, and an experiment in which they agree cannot say which the estimator reports. Driving mode $l$ at amplitude $q^l$ separates them. The orbit still closes onto an $r$-torus, so $d_{\text{act}} = r$, while the effective rank falls to $(\sum_l a_l^2)^2 / \sum_l a_l^4$. The estimate follows the count: at four active directions the rank falls from 4.00 to 1.65 as $q$ goes from 1 to 0.5 while the estimate stays between 3.82 and 4.06, and over the grid it is closer to the count than to the rank, 1.02 against 1.67 (Appendix E). It counts only what it can resolve, as section 3.1 defines it. At eight directions with the weakest mode under a hundredth of the strongest, the estimate falls with the rank from 8.88 to 4.86: the torus it can see has fewer dimensions than the one built.

**The diagnostics.** With no ground truth, the identifiability ratio separates the accurate regimes, 1.00 to 1.09, from the inaccurate, 1.32 to 1.57. Under the slow drive it separates within a regime too, reading 0.99 where the estimate is correct and 1.52 where it has saturated. It is not sufficient alone: it reads 1.00 on the transient and must be paired with the trend-crossing count, near zero there. Both are calibrated on these five arms only. The degeneracy flag fires in a majority of windows on eleven of the thirty-five cells of section 7.

## 7 APPLICATION TO GROKKING

Grokking (Power et al., 2022) is studied in two settings that differ in every relevant respect, and we work in both (Appendix O lists every run and every departure from the source configurations). The first is the one-layer transformer of Nanda et al. (2023) with AdamW and cross-entropy, at weight decay 1.0 on modular addition and 0.2 on the $S_5$ composition task of Power et al. (2022), and on mini-batches of 256 where the source is full-batch, which is what makes it the stochastic setting of table 1. The second is the unregularised full-batch gradient descent of Gromov (2023), a two-layer quadratic perceptron on mean squared error, run on the modular polynomials of Doshi et al. (2024), which supply label-matched target pairs (Appendix M); we take their tasks and representability criterion, not their optimiser. The diagnostics refuse both settings' logs, so we pay for the trajectory instead: that is the order below.

### 7.1 THE DIAGNOSTICS REFUSE BOTH SETTINGS

The frozen configuration of section 5.4, with only the window and stride shortened to fit these records, to a third of each and $1\,000$, gives table 12 and figure 6. The regularised stochastic runs sit in a band of the identifiability ratio, $1.29$ to $1.60$, that the deterministic regimes never reach, and their estimates span a factor of four across the five of them: set by $E_{\max}$ and the record, not by a rank, which is what figure 1 predicts for stochastic excitation. The full-batch runs carry the signature of a transient instead: identifiability ratio unity, roughness of one part in a thousand, two crossings per window, delay participation ratio exactly one. In neither is the value a count.

A change inside a run might still be read: the configuration is constant within a run — $E_{\max}$, the lag, the window, the observer — and with it the level offset, so a change cannot be attributed to it. The floor such a change must clear is the observer's own drift, which section 6.3 bounds at $1.16$ components. We cannot test that here, for want of resolution rather than of signal: the frozen window spans $39\,990$ optimiser steps, and the two largest excursions in the set are nowhere near a transition (Appendix L). These logs also offer one observer, the parameter norm, and it is the one for which section 6.3 cannot exclude a smoothness explanation.

### 7.2 LABEL-MATCHED PAIRS

The perceptron setting supplies the control the first cannot: both members of a pair fit the training set completely under identical optimisation and differ by one monomial, so anything separating their logs belongs to the task–solution pair. The median estimate is about three units higher for the non-generalising member in all four pairs, with no overlap between the two groups of medians, though the window-level distributions do overlap. It is not a dimension: all eight runs are transient, the separation is already visible in the raw loss curves (Appendix G), and the direct measurement does not separate them (Appendix J).

A fall is not specific to generalisation either. One occurs in a $200\,000$-step run at $p = 211$ with no weight decay that never leaves chance accuracy, and it is slow, of the kind section 7.4 attributes to the norm. The logs support only the weaker claim, that a fall is not evidence of generalisation (Appendix K).

### 7.3 THE TRAJECTORY SIMPLIFIES AT GENERALISATION

A stored trajectory delivers what a log cannot: the effective rank of section 3.1 is defined whether or not the orbit recurs. We sketch it in parameter and in function space at every logged step of a transformer trained to grok and report $\mathrm{PR}^{\mathrm{det}}$ (Appendix I). Near generalisation it collapses (figure 2): in function space by a factor of $5.2$ to $12.2$, median $7.0$, reaching bottom $235$ to $712$ steps after the generalisation step, and in parameter space to between $1.09$ and $1.60$, $150$ to $800$ steps later again. Both re-expand, in parameter space far past the plateau they left and in function space back to about it (table 13). Three things support the reading. It sits at generalisation in seeds whose generalisation step differs by a factor of $3.7$ and in a second task, so it does not track training progress; the displacement reaches its floor four thousand steps later and never recovers while the rank does, so the trajectory has not simply stopped; and the two weight-decay-zero runs fall by $1.6$ and $1.3$ in the same window.

Over their whole budget those controls fall by $3.2$ and $7.1$, the second inside the range of the generalising runs, so the groups differ in the timing and shape of the fall and not in its depth (section 7.4). None of this is $d_{\mathrm{act}}$, which a transient lacks: what concentrates on one direction is the variation inside the window, not the orbit.

### 7.4 WITHOUT WEIGHT DECAY THE FUNCTION SETTLES AND THE NORM DOES NOT

The two weight-decay-zero controls never leave chance validation accuracy, and show how little a low rank means alone. Both peak early, at about 33 in function space, then settle to $6.6$ in modular

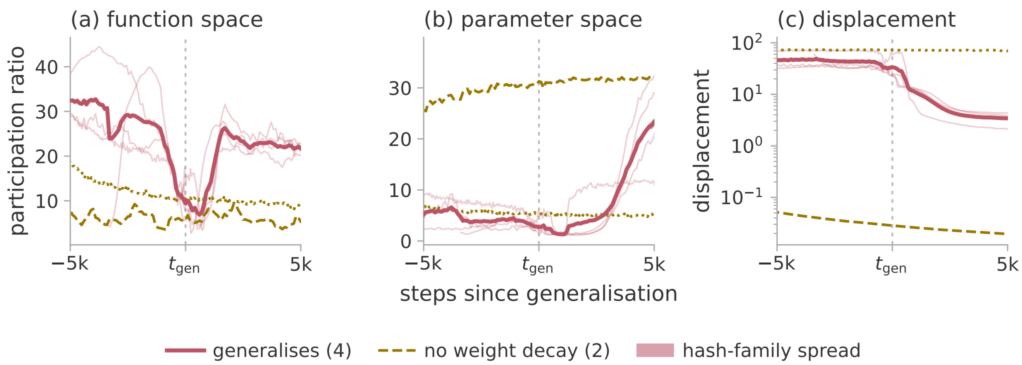


Figure 2: The trajectory effective rank measured directly, on three seeds of modular addition at weight decay $1.0$ and one $S_5$ composition at $0.2$, against the same two configurations at weight decay zero, which memorise and stay at chance validation accuracy. Colour is the regime of table 1, so the four regularised runs are stochastic and the two controls transient; the thick line is the mean of the four that generalise. The statistic is $\mathrm{PR}^{\det}$, an effective rank: these runs have no recurrent set, so no active dimension is defined for them. Windows are $600$ optimiser steps for the modular runs and $300$ for $S_5$, at their midpoint, so a feature is located to within $\pm 300$ and $\pm 150$ steps, and each control is aligned on the generalisation step of its matched run. The band on (a) and (b) is the two-hash disagreement, under two per cent of the mean (Appendix I).

addition and $8.1$ in $S_5$. Their parameter norm never comes back down, ending at $1.7$ and $2.6$ times its initial value, while every generalising run peaks and gives back a third to half of its growth. A low rank is therefore reached both by a run that has stopped moving and by one about to generalise, and only the shape of the fall separates them: the generalising runs hold a high rank through the plateau, collapse within a few hundred steps of the transition and recover, while the controls decline once and stay down. Their growing norm is also the ramped gain of section 6.3, the mechanism behind the scalar-log fall of section 7.2.

**Three rules follow.** Check the two diagnostics before reading a dimension off a log: both standard settings fail them. Measure the trajectory rank directly instead, for which a CountSketch to $1024$ dimensions suffices (Appendix I). And control it for window length and batch size first, since Appendix J and Appendix L show both move it by an order of magnitude.

## 8 Conclusion

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the independent components of the set the optimiser visits, to a mean absolute error of $0.87$ components up to eight of them. Outside that regime the set does not exist and what is returned is not a dimension. What makes it usable is the pair of diagnostics beside it, which flag those regimes with no ground truth. Applied to grokking they place both standard settings outside the admissible regime; measured directly instead, the trajectory effective rank collapses by a median factor of seven near generalisation and re-expands, in four regularised runs and much less in their controls.

**What this does not establish.** The count needs a recurrent regime, which none of the training runs we examined is in, and a lag matched to a period a log does not reveal. It is ordinal above eight components. The configuration is a choice as much as a measurement: the selection error spans a factor of five across the grid on identical data (Appendix C). The collapse rests on four runs against two controls that differ in the variable which also drives the parameter norm, is located only to within half a window, is a signature and not an early warning, and does not reproduce in the second setting (Appendix J). And the diagnostics are calibrated on five arms of one system. Three experiments would settle most of this: sweeping $E_{\max}$ and the record length to fix the ceiling; re-running the log estimate at a window matched to the transition (Appendix L); and descent above the stability threshold (Cohen et al., 2021), to see whether an undriven run is ever admissible.

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## A THE ESTIMATOR

**Algorithm 1** Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

**Require:** series $x_{1:n}$; lag $\tau$; embedding dimension $E$; neighbours $m$; Theiler exclusion $W_T$; dither $\sigma$

$z \leftarrow (x - \operatorname{mean} x)/\operatorname{std} x$; $\quad z \leftarrow z + \sigma\epsilon$, $\epsilon_t \sim \mathcal{N}(0, 1)$ ▷ breaks exact ties

$\boldsymbol{y}_t \leftarrow (z_t, z_{t-\tau}, \ldots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \ldots, n$

$S \leftarrow 0$; $N \leftarrow 0$; $n_{\text{dist}} \leftarrow 0$; $n_{\text{ratio}} \leftarrow 0$

**for** each $\boldsymbol{y}_i$ **do**

  $r_1 \le \cdots \le r_m \leftarrow$ distances to the $m$ nearest $\boldsymbol{y}_j$ with $|i - j| > W_T$

  $r_j \leftarrow \max(r_j, \varepsilon_{\text{dist}})$; count those clipped into $n_{\text{dist}}$

  $S_i \leftarrow \max\left(\sum_{j=1}^{m-1} \log(r_m/r_j),\ \varepsilon_{\text{ratio}}\right)$; count clips into $n_{\text{ratio}}$

  $S \leftarrow S + S_i$; $\quad N \leftarrow N + 1$

**end for**

flag the window degenerate if $n_{\text{dist}}$ or $n_{\text{ratio}}$ exceeds $1\,\%$ of its denominator

**return** $\big(N(m-1) - 1\big)/S$

Three details of the implementation are not cosmetic. The returned value is $(N(m-1)-1)/S$ rather than $N(m-1)/S$, which is the exactly unbiased pooled maximum likelihood estimate for a $\Gamma(N(m-1), 1/d)$ sample and differs from equation (4) by a factor $1 - 1/(N(m-1))$, below $10^{-5}$ at the sizes used here. A flagged window is still returned rather than discarded. The ladder pipelines drop flagged windows when a series is summarised over its sliding windows; the two grokking pipelines take the median over all of them, which changes nothing there because no published cell of table 12 contains a flagged window. And no clamping of the returned value to $E$ is applied, so a divergent estimate is reported as divergent: clamping would convert the transient regime of table 1 into a plausible number.

The floors are $\varepsilon_{\text{dist}} = 10^{-8}$ on a neighbour distance and $\varepsilon_{\text{ratio}} = 10^{-5}$ on the per-point log-ratio sum, with $\sigma = 10^{-9}$ on the standardised series. The Theiler exclusion is $W_T = \max\big((E-1)\tau,\, t_{\text{acf}}\big)$, where $t_{\text{acf}}$ is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples because the neighbour query cost is linear in it. Appendix N measures what that cap costs.

# B The observers

Twelve scalar observers are recorded on the system of section 5.4, in five families. Table 3 defines each. Two are excluded from every aggregate, for the reasons appendix D gives. “Parameter norms (two)” in table 5 is the pair $\|\boldsymbol{\theta}\|$ and $\|\boldsymbol{c}\|$, reported as one row because their errors agree to two decimals; they are distinct quantities and both are given here.

Table 3: The twelve observers. $\boldsymbol{c}$ is the coordinate in the $k$-dimensional subspace and $\boldsymbol{\theta}_0$ the frozen head at initialisation; $\boldsymbol{L}$ is the matrix of probe logits, the probe being a fixed held-out set drawn once before training; $\boldsymbol{g} = \partial\ell/\partial\boldsymbol{c}$. The projection directions $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{a}$, and the rotation $\boldsymbol{R}$, are drawn once from a fixed seed and never redrawn.

| name | family | definition |
|---|---|---|
| instantaneous loss | loss | the loss on the current minibatch |
| full-batch loss | loss | the loss on the whole training set |
| probe loss | loss | cross-entropy on the probe set |
| parameter norm | norm | $(\|\boldsymbol{\theta}_0\|^2 + \|\boldsymbol{c}\|^2)^{1/2}$ |
| subspace norm | norm | $\|\boldsymbol{c}\|_2$ |
| function-space norm | norm | $\|\boldsymbol{L}\|_F$ |
| gradient norm | gradient | $\|\boldsymbol{g}\|_2$ |
| gradient projection | gradient | $\langle \boldsymbol{u}, \boldsymbol{g} \rangle$ |
| fixed parameter projection | projection | $\langle \boldsymbol{v}, \boldsymbol{R}\boldsymbol{c} \rangle$ |
| function-space projection | function | $\langle \boldsymbol{a}, \operatorname{vec} \boldsymbol{L} \rangle$ |
| margin | function | mean probe logit gap, true class less best other |
| probe accuracy | function | accuracy on the probe set |

# C Frozen configurations

Two estimator configurations are used in this paper, one for each range of the active dimension. Both were selected once, on data withheld from every later experiment, and then frozen. Table 4 states them and the grids they came from.

Table 4: The two frozen configurations and their selection. Only the swept parameters were free; the remainder were fixed before any estimate was computed. Three settings are common to both and are not repeated: $m = 20$ neighbours, the autocorrelation Theiler rule of appendix A, and a dither of $10^{-9}$; the first two were swept in the eight-direction grid and these are the values it selected.

| | eight-direction | twenty-direction |
|---|---|---|
| $E_{\max}$ | 20 | 40 |
| $\tau$ | 4 | 16 |
| window / stride | 8 000 / 2 000 | 8 000 / 4 000 |
| swept | $E_{\max}$, $\tau$, $m$, Theiler, window | $E_{\max}$, $\tau$ |
| grid size | 48 | 4 |
| selection seeds | 90, 91, 92 | 0, 1, 2 |
| selection ranks | 2, 4, 6 | 2, 6, 10, 14, 18 |
| withheld seeds | 0–3 | none |
| withheld ranks | 1, 3, 5, 8 | the other fifteen |
| selection error, best | 0.32 | 2.47 |
| selection error, worst | 1.50 | 3.35 |

The selection objective in both cases is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection, one from each observer family. The eight-direction objective adds a penalty on the rank correlation and on the spread across seeds; at the twenty-direction configuration both penalties are identically zero, so the objective there reduces to the error alone.

Four properties of these choices should be stated with them.

*Both sit at a grid boundary.* The eight-direction configuration is at the top of all five swept axes and the twenty-direction one at the top of both. That is less alarming than it sounds for the first, since four of its five axes have only two levels, so every configuration in that grid is at a boundary in four of them; $\tau$ is the only axis with an interior value, and the winner is at its top end. It does mean neither configuration is interior, and that a wider grid could move the level.

*The selection error spans a factor of five on identical data.* Across the eight-direction grid the error runs from $0.32$ to $1.50$ on the same logs, so the level is a choice as much as a measurement, and only the withheld numbers of section 5.4 may be read as recovery.

*The Theiler axis is inert on this data.* The autocorrelation rule returns the larger of the embedding span $(E_{\max} - 1)\tau$ and the measured autocorrelation time. On these logs the autocorrelation time is one to three samples while the embedding span is nine or more, so the two settings return bit-identical values at all 24 points of the grid where they are compared. The 48-point grid therefore contains $24$ distinct measurements, and the choice of Theiler rule is not a choice at all here.

*Neither configuration always uses the window its parameters name.* At $E_{\max} = 40$ and $\tau = 16$ the requested exclusion is $624$ samples, which the implementation truncates to its cap of $150$ for the cost reason given in appendix A. The eight-direction configuration requests $76$ and is uncapped on every system of section 5, but on the training logs of section 7 the autocorrelation branch of the rule asks for $161$ to $858$ and is likewise truncated to $150$. appendix N measures what the truncation costs.

## D Results by observer, and the alternatives

Two rows of table 2 predate the estimator of appendix A and are marked in that table with the requirements they fail. The oscillating matrix was scored with an earlier implementation that applies no Theiler exclusion and clamps the returned value to $E$, at $E = 25$ and $m = 30$ rather than at the frozen configuration, and its grid swept the drive period alongside the estimator parameters, which is the breach of requirement 2 that also disqualifies the two regression systems; it reports no companion statistics, which is requirement 5. The function-subspace row shares the clamping and the mixed grid, computes one window per run rather than a median over sliding windows, and has no zero-learning-rate control, so requirement 4 is untested there rather than passed. Both rows are kept because they locate the ceiling and because the oscillating matrix supplies the synchronisation control of section 5.1, which no other system provides; neither should be read as a recovery result.

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.4, in the deterministic recurrent regime, and are scored against the measured effective rank rather than against $r$. Where an aggregation over seeds is needed, the median over the four seeds at each rank is taken before scoring; that is the convention the analysis uses throughout, and it is stated because the choice moves the numbers. Three figures in this paper describe the same eight-direction measurement under three aggregations and should not be read as different results: $0.87$ is the mean error run by run, $0.47$ the median over observers of each observer's error, and $0.457$ (table 7) the error of the single series left after taking the median over seeds and observers at each rank. Table 5 gives the result for every observer.

Table 5: Per-observer results. One row is one observer, except that the two parameter norms share a row; MAE and slope are over the four withheld ranks $r \in \{1, 3, 5, 8\}$ after taking the median over the four withheld seeds at each rank, scored against the measured effective rank. The rank correlation over those four ranks is $+1.00$ for every observer, so it has no column. The last column flags the observers whose roughness statistic orders the four withheld ranks perfectly: with $n = 4$ a rank correlation takes only eleven values and nothing short of $\pm 1$ is distinguishable from no ordering, so it is a flag and not a number.

| observer | family | MAE | slope | roughness orders rank |
|---|---|---|---|---|
| probe loss | loss | 0.30 | 0.90 | no |
| function-space norm | norm | 0.36 | 0.93 | no |
| margin | function | 0.41 | 0.93 | no |
| parameter norms (two) | norm | 0.45 | 0.85 | **yes** |
| fixed parameter projection | projection | 0.49 | 0.97 | no |
| gradient projection | gradient | 0.56 | 0.89 | no |
| function-space projection | function | 0.70 | 0.70 | no |
| gradient norm | gradient | 1.97 | 1.65 | no |
| full-batch loss | loss | 2.01 | 1.62 | no |
| instantaneous loss | loss | 0.42 | 0.86 | no |
| probe accuracy | function | degenerate in every window | | |

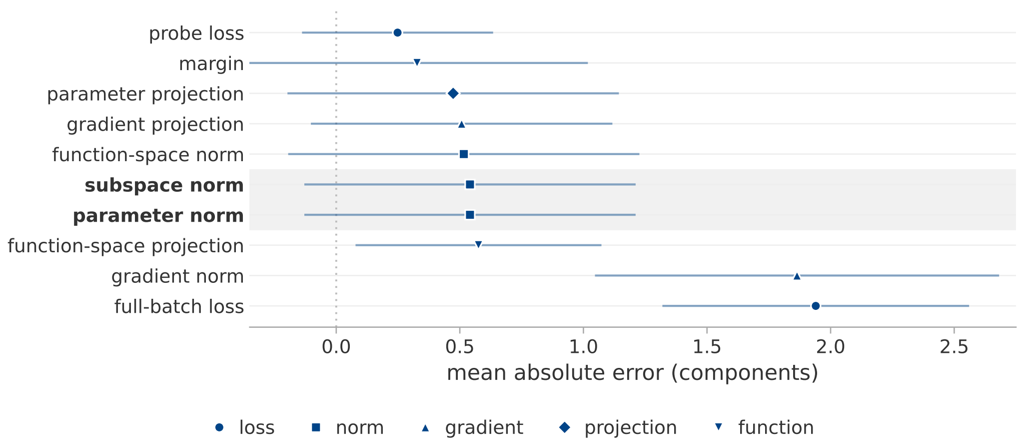


Figure 3: Which scalar log to keep. Ten of the twelve observers of table 3, scored on the fast recurrent arm of section 5.4. The point is the mean absolute error of the estimate against the measured effective rank over the seven simulated ranks, taking the median over the four seeds at each; the bar is the mean across-seed standard deviation of the estimate, drawn either side of the point. It is the spread of the estimate and not of the error, so it may extend below zero, and where it does the observer's error is inside its own seed-to-seed noise. Marker shape is the observer family. The shaded rows are the two parameter norms, which are what a training run ordinarily logs and the only two whose roughness also orders the withheld ranks of table 5. Eight of the ten lie between $0.25$ and $0.58$ components with every bar overlapping every other, so this experiment does not choose among them; the gradient norm and the full-batch loss are more than three times worse than the next observer and are the two to avoid. The ordering differs slightly from table 5, which scores only the four withheld ranks and reports the two parameter norms as one row.

Two observers are excluded from every aggregate, for different reasons. Probe accuracy is quantised, which drives the delay vectors together and trips the degeneracy flag in $100\,\%$ of its windows; it is the sole source of the degenerate fraction reported anywhere in this study, every other observer being at zero. The instantaneous mini-batch loss is excluded because it fails requirement 4: at zero learning rate, where the parameters do not move at all, its estimate still reads $0.91$, $2.22$, $3.71$, $4.62$, $5.45$, $5.31$ and $6.93$ across the seven simulated ranks $r = 1, \ldots, 6$ and $8$, a range of six components produced entirely by the drive. The other eleven observers are exactly constant under that control, so no estimate is defined for them.

The roughness column carries a caveat that the main text states and that this table makes precise. For eight of the nine scored rows the roughness statistic does not order the four withheld ranks, so their agreement with the truth is not available from smoothness. For the two parameter norms it orders the withheld ranks perfectly, and those are exactly the observers a training run ordinarily logs. Their result should therefore be read as consistent with, rather than as evidence against, a smoothness explanation.

Table 6: Twenty available directions, image data, median over three seeds and eight observers. The truth is the measured effective rank; the last row is the spread of MG across seeds at fixed rank.

| true $r$ | 1 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 17 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| MG | 0.89 | 2.35 | 4.41 | 6.66 | 7.38 | 8.99 | 10.61 | 11.40 | 13.50 | 15.09 |
| $\mathrm{PR}_{\mathrm{delay}}$ | 2.00 | 2.29 | 4.54 | 6.81 | 7.95 | 9.20 | 10.79 | 12.15 | 11.71 | 13.51 |
| spectral PR | 1.01 | 1.25 | 2.58 | 3.68 | 4.18 | 5.48 | 5.98 | 6.53 | 8.11 | 8.38 |
| spread across seeds | 0.00 | 0.20 | 1.49 | 2.15 | 2.11 | 2.95 | 1.50 | 2.84 | 2.76 | 3.33 |

Table 7: The estimator against the alternatives, on the same data and under the same convention, at both ranges. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum.

| | eight directions | | twenty directions | |
|---|---|---|---|---|
| statistic | MAE | $\rho$ | MAE | $\rho$ |
| MG † | 0.457 | +1.00 | 1.62 | +0.97 |
| LB † | 0.44 | +1.00 | 1.36 | +0.98 |
| TwoNN † | 1.81 | +1.00 | 3.48 | +0.89 |
| $\mathrm{PR}_{\mathrm{delay}}$ | 0.89 | +1.00 | 2.19 | +0.98 |
| spectral $\mathrm{PR}$, 256 bands | 1.60 | +1.00 | 5.29 | +0.99 |
| spectral $\mathrm{PR}$, 1024 bands | 1.35 | +1.00 | 4.47 | +0.99 |
| spectral $\mathrm{PR}$, native resolution | 1.03 | +0.80 | 2.78 | +0.93 |
| roughness | 3.65 | +0.20 | 9.90 | −0.57 |

Table 7 should be read with three things in mind, and none of them favours MG.

First, MG is not the most accurate statistic at either range. The uncorrected per-point average is ahead of it at both, by $0.02$ components at eight directions and by $0.27$ at twenty. The correction of section 3.2 removes a positive bias, and on a saturating estimate that bias was partly cancelling the saturation, so removing it costs accuracy exactly where the estimate is already saturating.

Second, MG is not the best-ordering statistic at twenty directions. The spectral participation ratio at $256$ and $1024$ bands orders the ranks more faithfully than MG does, at three times the error. If only the ordering is wanted, a statistic requiring no embedding and no neighbour search will supply it.

Third, MG's advantage over the linear delay participation ratio is a factor of about two at eight directions, not the factor of six that the selection split of appendix C reports. That larger figure is computed on the ranks and seeds the configuration was chosen on and is a property of the selection, not a recovery result.

**The advantage depends on the observer.** Running the same torus through two scalar observers that differ only in how evenly they weight the modes settles which statistic is preferable, and the answer is that it depends. Under an observer that weights all modes equally, the spectral participation ratio at $256$ bands attains an error of $0.35$ against MG's $1.19$, ahead by $3.4$ times. Under a generic observer with random weights, which is what a random projection of a network's parameters actually is, MG attains $0.34$ against $1.61$, ahead by $4.7$ times. Both orderings are perfect in three of those four cells, so what separates the statistics is absolute recovery and not rank agreement. Both figures also use the delay lag that is best for that observer, which a controlled experiment can choose and a training log cannot. Held at a lag common to both observers the two factors are $6.6$ and $4.7$ at $\tau = 4$, and $3.4$ and $1.9$ at $\tau = 1$: the direction of each advantage survives, its size does not.

## E CONDITIONS OF VALIDITY IN DETAIL

**The delay lag.** The delay window spans $(E_{\max} - 1)\tau$ samples, and table 8 sweeps $\tau$ on a fixed torus of period $400$ at $E_{\max} = 20$, leaving the system unchanged. At a span of one twentieth of a period every rank returns the same tangent constant near $2.5$; at four tenths it follows the rank, to within one component at six and eight directions and overshooting by one to two at three and four; at one and a half periods it exceeds the rank eightfold. The estimate at $r = 8$ moves by $60.7$ components across the sweep and the estimate at $r = 1$ by $0.03$, so the failure is not a uniform bias but a collapse of the ordering at one end and a divergence at the other.

Table 8: Estimate against the delay lag on a fixed torus of period 400, $E_{\max} = 20$. The span is the delay window in units of the oscillation period. The correct rank is the column heading.

| $\tau$ | span | $r = 1$ | 2 | 3 | 4 | 6 | 8 |
|---|---|---|---|---|---|---|---|
| 1 | 0.05 | 1.42 | 2.44 | 2.49 | 2.63 | 2.44 | 2.60 |
| 4 | 0.19 | 1.43 | 2.72 | 3.31 | 3.83 | 3.94 | 4.29 |
| 8 | 0.38 | 1.43 | 2.77 | 4.18 | 6.16 | 6.57 | 8.53 |
| 16 | 0.76 | 1.44 | 3.00 | 5.92 | 13.66 | 12.83 | 20.76 |
| 32 | 1.52 | 1.46 | 3.06 | 9.90 | 33.28 | 31.02 | 63.35 |

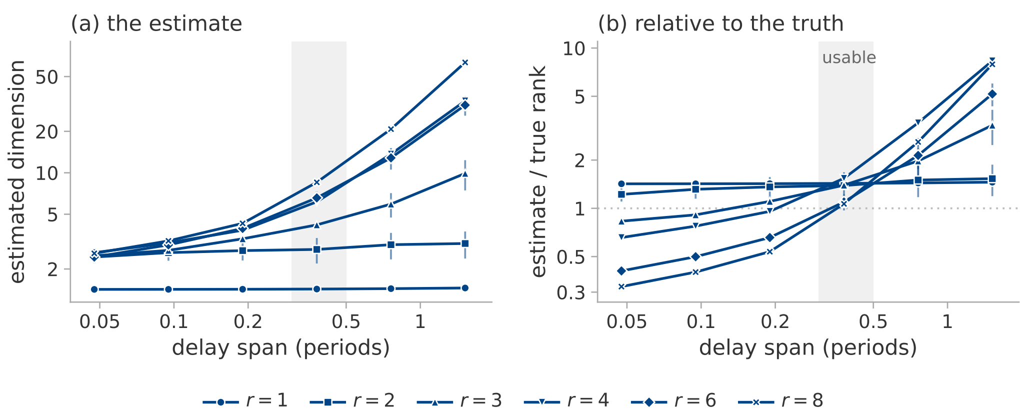


Figure 4: The delay lag, swept on one system: a fixed torus of period 400 samples at $E_{\max} = 20$, two seeds, six fixed lags $\tau \in \{1, 2, 4, 8, 16, 32\}$ spanning 0.05 to 1.52 periods. Marker shape is the true rank, the marker the median of the two seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period. Below a tenth of a period every rank returns the same tangent constant near 2.5 and the ordering is gone; above one period the estimate diverges, reaching 63.4 at $r = 8$; in between it is within one component at $r = 6$ and $r = 8$ and overshoots by one to two at $r = 3$ and $r = 4$. Across the sweep the estimate at $r = 8$ moves by 60.7 components and at $r = 1$ by 0.03, so the failure is a collapse of the ordering at one end and a divergence at the other, not a uniform bias. Table 8 gives the same numbers. A seventh setting, in which $\tau$ is taken from the autocorrelation, is omitted because its span differs between the two seeds.

**Nuisance factors.** Table 9 varies seven factors at a fixed active dimension of four, over five seeds and twelve observers, against the real four-component change measured below. Rotating the coordinates is scored on the fixed parameter projection alone, since the other eleven observers are invariant to it by construction, so it measures a property of the observer rather than of the estimator.

Table 9: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale against which the others should be read. The last column is a percentage of the 3.95-component fall that the change-detection experiment below observes for a true four-component change. A constant rescaling of the observer, which the scorer standardises away, moves the estimate by $0.000$ on every observer and is omitted.

| factor | $\lvert\Delta\mathrm{MG}\rvert$ | as % of a real four-component change |
|---|---|---|
| none (baseline) | 0.21 | 5 |
| learning rate halved | 0.16 | 4 |
| drive band up one octave | 0.49 | 12 |
| drive band down one octave | 0.50 | 13 |
| coordinates rotated | 0.86 | 22 |
| observer gain ramped $10\times$ | 1.16 | 29 |
| state amplitude ramped $4\times$ | 1.32 | 33 |

**Detecting a change.** Recovery of a static level does not establish that a change is detected. Running three consecutive segments $r_{\mathrm{high}} \to r_{\mathrm{low}} \to r_{\mathrm{high}}$ with nothing else altered, and re-measuring the ground truth on each segment, a true fall of $3.99$ components is observed as $3.95$ in the deterministic recurrent regime, is detected by every usable observer in both directions, and recovers to the level before it to within $0.01$ components. Under stochastic excitation the same true change is observed as $0.15$.

The rule fires when the estimate falls below the median of the pre-switch segment by three times that segment's window-to-window scatter, for two consecutive windows. The threshold uses the pre-switch segment alone, never the midpoint of the two observed levels, which would hand the detector the answer and would decide the arm where the two levels coincide by a coin flip. On the parameter norm the level pairs read $4.77 \to 1.22 \to 4.78$ for a true $4 \to 1 \to 4$ and $6.90 \to 2.12 \to 6.91$ for a true $6 \to 2 \to 6$; at $8 \to 3 \to 8$ they read $6.97 \to 3.64 \to 6.97$, under-reporting the high level by a component, which is the saturation of table 6 seen from another direction. The level error is $0.89$ at the high level and $0.22$ at the low one: the estimate is more accurate the fewer directions are active.

The response time cannot be quoted from this experiment. With a window of $8\,000$ samples and a stride of $2\,000$ the two observed lags, 999 and 1 999 steps, are the two floors of the measurement grid rather than measured values. What the experiment does establish is that the estimate crosses the threshold on a window still containing $87.5\,\%$ pre-switch data.

**Anisotropy.** Table 10 gives the measurement of section 6.4 in full: the system of section 5.1 with mode $l$ driven at amplitude $q^l$, three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration. The manifold dimension is $r$ in every row. The measured participation ratio agrees with the closed form $(\sum_l a_l^2)^2/\sum_l a_l^4$ to two decimals throughout, which is the check that the construction does what it claims.

Table 10: An $r$-torus with geometrically decaying amplitudes. $r$ is the manifold dimension and is constant along each block, $q$ the amplitude decay factor; PR is the participation ratio of the covariance of the $r$ oscillating coordinates. Median over three seeds.

| $r$ | $q$ | PR | MG | $\mathrm{PR}_{\mathrm{delay}}$ |
|---|---|---|---|---|
| 2 | 1.0 | 2.00 | 2.30 | 3.12 |
| 2 | 0.7 | 1.79 | 2.33 | 3.28 |
| 2 | 0.5 | 1.47 | 2.35 | 3.18 |
| 4 | 1.0 | 4.00 | 3.82 | 6.44 |
| 4 | 0.7 | 2.60 | 3.89 | 4.85 |
| 4 | 0.5 | 1.65 | 4.06 | 3.11 |
| 6 | 1.0 | 6.00 | 8.76 | 8.23 |
| 6 | 0.7 | 2.84 | 6.90 | 5.53 |
| 6 | 0.5 | 1.67 | 4.83 | 3.21 |
| 8 | 1.0 | 8.00 | 8.88 | 8.76 |
| 8 | 0.7 | 2.90 | 5.26 | 5.19 |
| 8 | 0.5 | 1.67 | 4.86 | 3.00 |

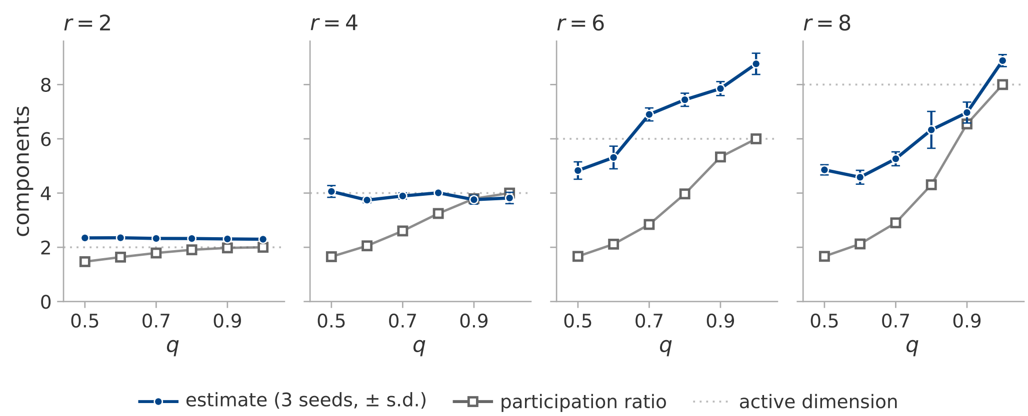


Figure 5: Anisotropy separates the two candidate estimands, and the estimate follows the count. The system of section 5.1 with mode $l$ driven at amplitude $q^l$, three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration; one panel per $r$. The manifold dimension is $r$ in every panel, drawn dotted: the orbit closes onto an $r$-torus whatever $q$ does. Filled circles are the estimate, the median over seeds, with their standard deviation as bars; open squares are the measured participation ratio of the covariance, which agrees with the closed form $(\sum_l a_l^2)^2/\sum_l a_l^4$ to two decimals in all twenty-four cells and so carries no spread. At $r=2$ and $r=4$ the estimate is flat in $q$ while the participation ratio falls to $1.47$ and $1.65$: the estimator counts the coordinates of the set rather than measuring its effective rank. At $r=6$ and $r=8$ it falls too, once the weakest mode is under a hundredth of the strongest, because it counts only the directions it can resolve, as section 3.1 defines it. Table 10 gives twelve of these points in digits.

At two and four directions the estimate is flat in $q$ while the participation ratio falls by a factor of $1.4$ and $2.4$, which is the clean case. At six and eight it falls too, and the reason is visible in the amplitudes: at $q=0.5$ and $r=8$ the weakest mode is driven at $0.008$ of the strongest, far below the neighbourhood scale the estimator works at, so the torus it sees really does have fewer dimensions than the one that was built. The estimator is therefore counting modes it can resolve, which is neither the manifold dimension of the construction nor the participation ratio of its covariance, and coincides with both only when the amplitudes are equal.

# F SYSTEM CONSTRUCTIONS

**The image-data system of section 5.4.** The data is partitioned into twelve fixed groups, the same partition at every rank so that the partition is not itself a function of $r$, and the loss weights of $r$ of them are modulated by $r$ rationally independent sinusoids filling one octave. Modulating group $j$ tilts the gradient along a direction $\phi_j$. These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\mathrm{pinv}(\Phi)Q$ with $Q$ an orthonormal basis of $\mathrm{range}(\Phi)$ orthogonalises them, and the per-mode scalar response gain of the linearised update is divided out. Without this correction the trajectory participation ratio falls well below $r$.

Table 11 verifies that the construction achieves what it intends, and two of its rows are results rather than settings. A decaying transient has no recurrent set whatever $r$ is, since a converging trajectory is a curve; and plain mini-batch descent sits at $6.2$ regardless of $r$, because its noise rank is fixed by the per-example gradient covariance of the data and is not a free parameter of the experiment. Projecting real mini-batch noise onto $r$ directions recovers only part of the intended rank, reaching $5.6$ at $r = 8$, which is why that regime is scored against its measured value and not against $r$.

Table 11: Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.4. Every estimate in this paper is scored against these values, never against $r$.

| regime | $r = 1$ | 2 | 4 | 6 | 8 | what is excited |
|---|---|---|---|---|---|---|
| `qp` | 1.00 | 2.00 | 4.00 | 6.00 | 8.00 | $r$ sinusoids, fast drive |
| `qp_slow` | 1.00 | 2.00 | 4.00 | 6.00 | 7.99 | the same at a training-log timescale |
| `qp_nopre` | 1.00 | 1.99 | 3.99 | 5.89 | 7.55 | `qp` without the preconditioner |
| `mixed` | 1.00 | 2.00 | 4.00 | 6.00 | 7.99 | `qp` plus noise in the same directions |
| `noise` | 1.00 | 2.00 | 3.99 | 5.99 | 7.98 | injected rank-$r$ gradient noise |
| `noise_nopre` | 1.10 | 2.08 | 3.74 | 5.47 | 6.43 | the same without the preconditioner |
| `batch_proj` | 1.00 | 1.79 | 3.25 | 4.53 | 5.61 | real mini-batch noise projected to rank $r$ |
| `batch` | 6.20 | 6.25 | 6.21 | 6.26 | 6.20 | plain mini-batch descent |
| `gd` | 1.05 | 1.05 | 1.05 | 1.03 | 1.06 | a decaying transient |
| `qp`, $\eta = 0$ | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | requirement 4 control |

The last row is the control of requirement 4. With the learning rate multiplied by zero the active dimension is identically one at every nominal rank, so an observer whose estimate still tracks $r$ there is reading the drive rather than the optimiser.

The functional rank is 10 at every rank, seed and regime, with a participation ratio between $8.40$ and $9.03$ across the four seeds; it is a property of the frozen backbone and does not depend on the excitation. The condition number of the drive-direction matrix is $1.00$ at $r = 1$ by construction and has a median of $13.3$ at $r = 8$, which is the anisotropy the mixing matrix exists to remove.

**The function-space construction.** A local adapter is built around a trained $64 \to 16 \to 10$ network and its function-space Jacobian is QR-whitened, so that its first $r$ columns have rank $r$ at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it by exact backpropagation through both layers. The functional, trajectory-covariance and update-covariance ranks were all verified to equal $r$.

**The twenty-direction system.** The same construction at twenty available directions reproduces the nominal rank to within $0.008$ components over $r = 1, \ldots, 20$ in the recurrent regime, and to three decimals up to $r = 10$. The other regimes behave as at ten directions: injected noise tracks $r$ closely, projected mini-batch noise reaches only $9.4$ at $r = 20$, and the decaying transient stays between $1.02$ and $1.09$ throughout.

## G DIAGNOSTICS AND FIGURES FOR THE GROKKING APPLICATION

Table 12: The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is the median over the run's sliding windows. Seven of the seventeen runs of table 18 are shown; figure 6 plots all seventeen, and the ranges quoted in section 7.1 are over all five regularised runs, not the three shown here. Runs that have not generalised within the budget are censored, not negative.

| run | generalises | MG | $\rho_{\text{ident}}$ | roughness | $\text{PR}_{\text{delay}}$ | crossings |
|---|---|---|---|---|---|---|
| transformer, positive control | yes | 15.31 | 1.29 | 0.030 | 1.02 | 56 |
| transformer, reduced data, s0 | yes | 4.93 | 1.60 | 0.068 | 1.52 | 146 |
| transformer, reduced data, s2 | no | 4.47 | 1.51 | 0.149 | 3.94 | 151 |
| transformer, no weight decay | no | 28.08 | 1.00 | $< 0.001$ | 1.00 | 2 |
| perceptron, $n + m$ | yes | 19.54 | 1.00 | 0.001 | 1.00 | 2 |
| perceptron, $n \cdot m$ | yes | 19.63 | 1.00 | 0.001 | 1.00 | 2 |
| perceptron, $n^3 + nm^2 + m$ | no | 22.43 | 1.00 | $< 0.001$ | 1.00 | 2 |

Figure 6 places every training log of section 7 in the plane of the two diagnostics, against the region in which every accurate case of section 6 fell. No run of either setting is in it: the regularised mini-batch runs are unstable in the embedding dimension, and the full-batch and zero-weight-decay runs are stable only because their series is monotone.

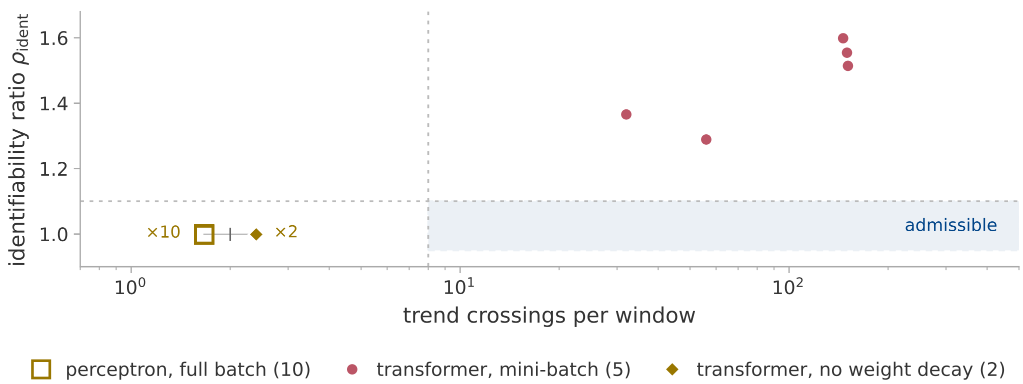


Figure 6: Every training log of section 7 in the plane of the two diagnostics, one point per run: seven transformer runs of $120\,000$ steps observed through the parameter norm, and ten perceptron runs of $100\,000$ steps observed through the training loss. Colour is the regime of table 1 and marker shape the setting: the five regularised mini-batch runs are stochastic, the ten full-batch perceptron runs and the two zero-weight-decay transformer runs transient. The vertical axis is the identifiability ratio, near unity when the estimate does not depend on the embedding dimension; the horizontal axis is the trend-crossing count, which tests non-monotonicity of the series and not recurrence in the reconstructed space. The grey lines are the edges of the shaded rectangle: to their left too few crossings, above them instability in $E$. Twelve of the seventeen runs coincide at exactly two crossings with $\rho_{\text{ident}}$ between $0.9975$ and $0.9992$; those two groups are drawn with a small offset either side of their true abscissa, which the grey tick marks, and labelled with their multiplicity. The shaded rectangle is where every accurate case of section 6 fell, $\rho_{\text{ident}} \leq 1.10$ and more than eight crossings. It describes the calibration and is not a validated decision rule; no run of either setting is inside it, and the two settings fail in different corners without reference to any label, the stochastic runs by having no invariant set and the transient ones by being monotone.

Figure 7 shows the separation of section 7.2 and the mechanism behind it. The median estimate is about three units higher for the member of each pair that never generalises, in all four pairs, though the window-level distributions overlap. The training loss shows that the two members differ in the shape of their curves long before any question of dimension arises.

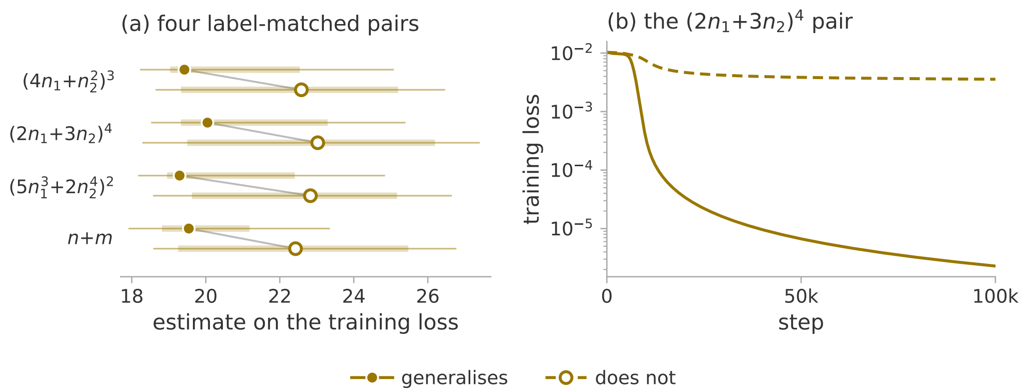


Figure 7: (a) The estimate on the training loss for the four label-matched pairs of section 7.2: three modular polynomials at $p = 97$, each against itself perturbed by one monomial, and modular addition against $n^3 + nm^2 + m$. Both members of a pair reach full training accuracy under identical optimisation on identical data budgets. All eight runs are full-batch perceptrons, which figure 6 places in the transient regime, so the figure carries that one hue and generalisation is carried by fill and dash. The marker is the median over the seven sliding windows, the thick bar their interquartile range and the thin line their full range. The medians separate the two members of every pair, by $2.88$ to $3.54$ units and with no overlap between the two groups of medians, but the window-level ranges overlap heavily, because the estimate drifts upward over training in every run rather than scattering about a level. (b) The $(2n_1{+}3n_2)^4$ pair, the second row of (a). The representable target drives the training loss to $2.3 \times 10^{-6}$ while the perturbed one stops at $3.6 \times 10^{-3}$, three orders of magnitude short despite fitting the training set completely, so the two curves differ in shape rather than in any attractor dimension. Across the four pairs that gap runs from 2.7 to 3.2 orders.

# H The collapse, run by run

Table 13 states the measurement of section 7.3 for every run. The plateau is the median of the $4\,000$ steps before the generalisation step, the dip the minimum within $\pm 4\,000$ of it, and "at" the position of that minimum relative to it; the controls, which have no generalisation step, are measured in the window of their matched weight-decay run. The last column is the largest value the statistic reaches after its dip, which is what "re-expands" means: in parameter space it exceeds the plateau by an order of magnitude in every run, in function space it returns to about the plateau and exceeds it in three of the four.

Table 13: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The two control rows in each block are measured in the aligned window; their deepest fall anywhere in their own budget is $3.15$ and $7.06$ in function space and $1.66$ and $2.37$ in parameter space, which is the comparison section 7.3 also reports.

| space | run | plateau | dip | depth | at | post-dip max |
|---|---|---|---|---|---|---|
| function | `mod_wd1` | 23.12 | 4.42 | 5.23 | +595 | 24.95 |
| | `mod_wd1_s43` | 33.06 | 2.71 | 12.21 | +235 | 27.82 |
| | `mod_wd1_s44` | 28.26 | 3.65 | 7.74 | +515 | 31.58 |
| | `s5_wd1` | 21.15 | 3.41 | 6.20 | +712 | 24.21 |
| | `mod_wd0` | 5.79 | 3.60 | 1.61 | — | — |
| | `s5_wd0` | 11.75 | 8.86 | 1.33 | — | — |
| parameter | `mod_wd1` | 3.09 | 1.09 | 2.84 | +1 095 | 35.34 |
| | `mod_wd1_s43` | 2.70 | 1.15 | 2.35 | +935 | 38.04 |
| | `mod_wd1_s44` | 2.71 | 1.13 | 2.39 | +1 315 | 37.34 |
| | `s5_wd1` | 7.00 | 1.60 | 4.38 | +862 | 13.43 |
| | `mod_wd0` | 30.10 | 27.70 | 1.09 | — | — |
| | `s5_wd0` | 5.59 | 4.53 | 1.24 | — | — |

# I The trajectory sketch

The direct measurement of section 7.3 needs the trajectory itself, which for the modular transformer is $226\,816$ doubles at each of two to three thousand logged steps ($228\,608$ for $S_5$). Every logged step is therefore compressed by a CountSketch (Charikar et al., 2002; Weinberger et al., 2009) to $\mathbb{R}^{1024}$, with two independent hash families. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length. The normalisation is not cosmetic. The raw logit scale is mechanically coupled to weight decay, so a signal read off it could not be distinguished from a detector of weight decay.

We do not verify a spectral guarantee for this. The elementary bound for a sketch of this kind is on pairwise inner products (Weinberger et al., 2009), and a participation ratio is a functional of the whole spectrum, on which an entrywise Gram bound is weak. A subspace-embedding guarantee does exist for CountSketch (Clarkson & Woodruff, 2013) and would apply here, since a sixty-sample window has rank at most sixty, but it requires a rank we do not measure per window. What the two hash families supply instead is a direct estimate of the error: every reported value is the mean of the two, and their disagreement is recorded with it. Over the published transformer windows that disagreement is $1.1\,\%$ of the value at the median and $8.4\,\%$ at its worst, which is the number the reader should use, and it is smaller than every effect section 7.3 reads.

Two further checks stand behind the numbers. Table 14 pushes synthetic trajectories of known rank through the sketch and compares against the same statistic computed on the uncompressed trajectory. The sketch is accurate to $0.04$ components throughout; the shortfall against the nominal rank at $5$ and $10$ is a property of the participation ratio on a sixty-sample cloud and is present without any compression, which is what separates the two questions. Nothing above rank ten is tested, and the plateaus in figure 2 are higher than that. On a real run of the perceptron architecture, where the parameter count is small enough to allow it, the participation ratio computed exactly on all $145\,500$ raw parameters agrees with the sketched value to three decimals. And the observer is verified not to perturb training by requiring the log to be bit-identical with and without it, over four hundred steps at reduced probe size, which matters because the task module seeds one global random stream that the train–validation split, the initialisation and the mini-batch order all continue: a single stray draw changes the initial weights and can destroy grokking.

Table 14: The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in 145 500 dimensions. The last column is what the compression costs; the gap between the first two is what a sixty-sample participation ratio costs, and is not attributable to the sketch.

| true rank | PR, uncompressed | PR, sketched | sketch − uncompressed |
|---|---|---|---|
| 1 | 1.00 | 1.00 | 0.00 |
| 2 | 1.99 | 1.98 | −0.01 |
| 5 | 4.42 | 4.46 | 0.04 |
| 10 | 8.61 | 8.63 | 0.02 |

Windows are sixty logged rows, which is 600 optimiser steps for the modular runs and 300 for $S_5$, and are labelled by their centre, so a feature at a given step is located to within half a window and the window is not causal. Five statistics are computed per window in each space, of which the paper reports two: $\mathrm{PR}^{\mathrm{det}}$, the participation ratio of the positions after removing a per-window linear trend, since a steady drift is rank one and would otherwise mask everything else; and $\mathrm{PR}^{\mathrm{upd}}$ on the increments block-averaged over five logged steps, to suppress the mini-batch noise floor. Beside them the total displacement is recorded, as the null for a rank that falls simply because the trajectory stopped moving.

## J The full-batch setting

Repeating the direct measurement on the perceptron setting gives a result that must be read as a limit of the measurement rather than as a negative. Under full-batch gradient descent every window of every run has a participation ratio between 1.000 and 1.362 on the four statistics the comparison uses, and between 1.000 and 2.171 over all eight recorded columns. This is not an artefact of the sketch: computed exactly on all 145 500 raw parameters the value is the same to three decimals, and the sketch recovers synthetic trajectories of known rank up to ten. A deterministic update makes the trajectory a smooth curve, and over a 600-step window a smooth curve is straight to within the measurement precision, so what the statistic reports is the curvature of the trajectory at the scale the window sets, and at that scale there is almost none. It has no range in which a collapse could appear, and the absence of a feature is therefore not evidence of its absence: the window must be set in units of the curvature scale of the trajectory. Repeating the measurement against window length confirms the diagnosis and withdraws the opening it appeared to leave. Two runs of 150 000 steps, a generalising one and its label-matched control, are measured over windows spanning 600 to 120 000 optimiser steps, with the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound on the statistic and its noise floor stay exactly where the published measurement put them and only the span changes. Table 15 gives the result. The statistic does leave the floor, at a scale of about $10^4$ steps, two orders of magnitude above the window that produced the 1.000–1.362: the absence of a feature there was a resolution limit and not a fact about the run.

It is not, however, a hidden signal. What appears at 12 000 steps is one localised maximum, and both runs have one, of the same size: 1.877 centred at step 12 800 for the run that memorises at 9 200 and generalises at 11 950, and 1.843 centred at 9 350 for the control, which memorises at 9 800 and does not generalise within the budget. Function space agrees with more range and no more separation, 2.618 against 2.145. What cannot be concluded from this is where the maximum sits relative to either milestone. The two centres are adjacent points of a 3 450-step grid and their windows overlap over 6 900 to 15 250 steps, so a 12 000-step window cannot localise a feature to better than the interval between the two milestones; measured against memorisation the offsets are +3 600 and −450, which is a difference the design cannot resolve. The claim the experiment does support is the negative one: at the window where the statistic has range, the generalising run and its label-matched control are not separated by it. Beyond 30 000 steps the ordering reverses and the control moves further than the generalising run, because the window is then a large fraction of a run whose weight norm is still climbing and the statistic is reading that drift. The full-batch trajectory therefore carries no rank signature of generalisation at any window length between 600 and 120 000 steps, which is the stronger form of the same conclusion: the collapse of section 7.3 belongs to the regularised stochastic setting, and a trajectory participation ratio must be controlled for window length as well as for batch size before anything is read from it.

Table 15: Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window. Sixty samples per window at every length. The 600-step row is the measurement reported above, on the same configuration and seed.

| | generalises | | control, never generalises | |
|---|---|---|---|---|
| window (steps) | median | max | median | max |
| 600 | 1.002 | 1.035 | 1.001 | 1.011 |
| 3 000 | 1.070 | 1.253 | 1.003 | 1.092 |
| 6 000 | 1.028 | 1.537 | 1.005 | 1.360 |
| 12 000 | 1.011 | 1.877 | 1.020 | 1.843 |
| 30 000 | 1.007 | 2.111 | 1.113 | 1.725 |
| 60 000 | 1.021 | 1.660 | 1.335 | 1.975 |
| 120 000 | 1.256 | 1.659 | 1.929 | 2.337 |

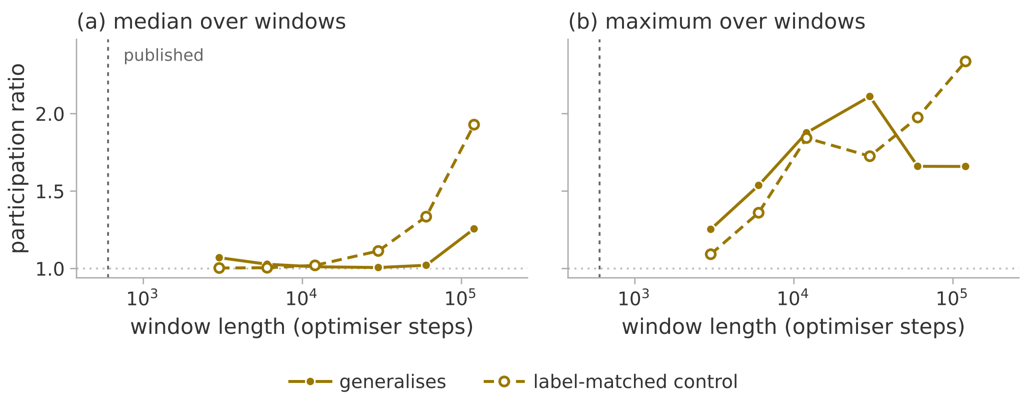


Figure 8: The trajectory participation ratio against window length, on the full-batch pair of this appendix: modular addition, which memorises at 9 200 and generalises at 11 950, and its label-matched control, which memorises at 9 800 and does not generalise. The statistic is $\mathrm{PR}^{\mathrm{det}}$ on the sketched parameter trajectory over windows of 3 000 to 120 000 optimiser steps, the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound and the noise floor stay where the published measurement put them and only the span changes. (a) The median over a run's windows, (b) the maximum; the dotted line at 1 is the floor a straight trajectory returns, and the dashed vertical the 600-step window of section 7.1, which is not part of this sweep. The median stays on the floor until about $10^4$ steps, two orders of magnitude above that window, so the absence of a feature there was a resolution limit and not a fact about the run. Where the statistic finally has range it does not separate the two runs, and beyond 30 000 steps the ordering reverses, because the window is then a large fraction of a run whose weight norm is still climbing. Both runs are transient, so the figure carries one hue and the control is drawn open and dashed.

Introducing mini-batches confirms the diagnosis. The same statistic then ranges over more than a decade while the generalisation step moves by one logged step, so the learning is unchanged and the statistic has moved by an order of magnitude. Any claim read from a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

## K What falls, and when

A fall of the estimate is not specific to generalisation. In a 200 000-step run at $p = 211$ with no weight decay, which memorises at step 1 800, ends at 3.6 % validation accuracy and grows its parameter norm from 43 to 126, a fall is detected 49 250 steps after memorisation, and the amplitude control of section 6.3 supplies a mechanism that needs no dimension at all: a ramped observer gain moves the estimate by 1.16 components, or 29 % of a real four-component change.

What that run cannot show is that its trajectory simplified. Its log sits in the transient regime of table 12, where the estimate is not a dimension, so a fall in it is a fall of a quantity we have argued should not be read. The two must be kept apart: the direct measurement of section 7.3 is a dimension and falls in four generalising runs and not in their controls, whereas the estimate on a log, in these settings, is not a dimension and falls in at least one run that does not generalise. What follows is the weaker and defensible claim, that a fall on a log is not evidence of generalisation. Nor is the absence of generalisation within a budget evidence that a fall was a false alarm: two reduced-data configurations against which earlier specificity figures were computed generalise once the budget is extended, at steps $39\,600$ and $110\,940$, so such runs are censored and no specificity figure may be computed against them.

## L WHAT A WINDOWED ESTIMATE ON A TRAINING LOG CAN RESOLVE

The claim of section 7.1 that a change within a run could be read against a fixed configuration is testable on the seven extended runs, and the test returns a resolution limit rather than an answer. The frozen configuration takes a window of a third of each record: $4\,000$ logged samples, which at the logging stride is $39\,990$ optimiser steps, advanced by $10\,000$ steps. That is nine windows per run and sixty-three weight-norm windows in all, and it localises a feature only to $\pm 19\,995$ steps, sixty-seven times coarser than the direct measurement of section 7.3.

The consequence is that the alignment figure 2 performs cannot be performed here. Of the twenty-seven window centres belonging to the three generalising runs, one lies within $5\,000$ steps of its own generalisation step and two within $10\,000$; the three runs' offset ranges are disjoint, so no offset has more than one run defined and no pooled mean exists. For the positive control the entire pre-generalisation phase is thirteen per cent of one window; for the reduced-data run that generalises at $110\,940$, no window centre falls after the transition at all.

What the runs do show is consistent with that. The one run with a centre near its generalisation step falls by $3.54$ components over the next stride, three times the floor — but the step lies inside four of its nine windows, and the falling windows are where its parameter norm is growing, which is the ramped gain of section 6.3. The other two show nothing: one has a total range of $0.77$ components over its whole record. Nor is any of this specific. Across the fifty-six stride-to-stride changes, six of twenty-four clear the floor in the runs that generalise and six of thirty-two in the runs that do not. The largest change in the set, $6.36$ components, is in a run that does generalise, but it falls more than $100\,000$ steps after its generalisation step; the next largest, $5.20$, is in a run that ends at seven per cent validation accuracy. The tightest comparison available is three runs of one configuration, of which one generalises: their ranges are $0.77$, $0.78$ and $0.67$ components, and nothing separates them.

This is the argument of appendix J in a second place. At the window the configuration provides, the statistic has no range in which a feature localised to the transition could appear, so its absence is not evidence of absence. Testing the claim needs a window set in units of the width of the transition, which for these runs is hundreds of steps and not tens of thousands; that is a re-run of the estimator, not a re-reading of what we have.

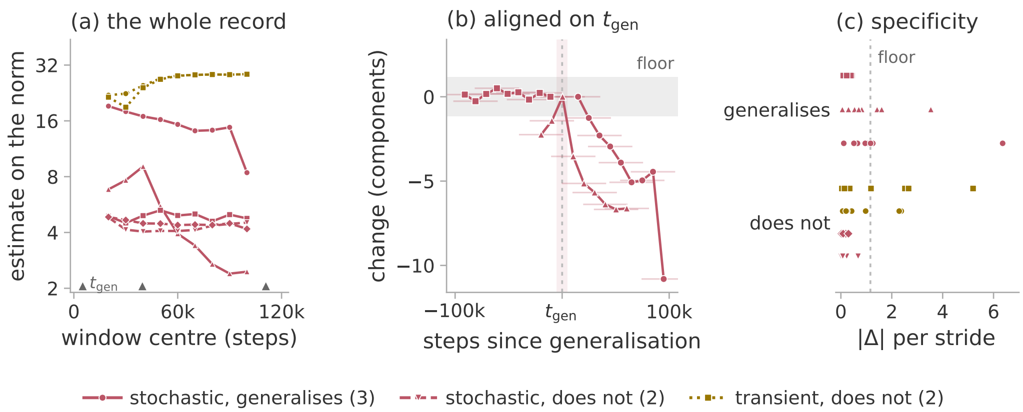


Figure 9: The windowed estimate on the parameter norm for the seven extended $120\,000$-step runs, under the frozen configuration of section 7.1: nine windows per run of $39\,990$ optimiser steps each, advanced by $10\,000$, plotted at their centres. Colour is the regime of table 1, five runs stochastic and two transient, none recurrent. (a) All sixty-three windows, with the three generalisation steps marked on the axis. (b) The three generalising runs aligned on their own generalisation step, each point a change from the window before it and its bar the span it summarises, so consecutive points share three quarters of their span and nothing is localised to better than $\pm 20\,000$ steps. The grey band is the $\pm 1.16$-component floor of section 6.3 and the shaded column is $\pm 5\,000$ steps of $t_{\text{gen}}$, which one of the twenty-seven centres reaches. (c) Every stride-to-stride change, one sub-row per run so that coincident changes stay countable, against the same floor. Six of the twenty-four changes in the runs that generalise clear it and six of the thirty-two in the runs that do not. The largest in the set, $6.36$ components, is in a run that generalises but falls more than $100\,000$ steps after its generalisation step; the next largest, $5.20$, is in a run that ends at seven per cent validation accuracy.

## M THE REPRESENTATION DIMENSION IN CLOSED FORM

The perceptron setting of section 7 makes the learned representation analytically available, and three different numbers can be read off it. They are not commensurable and the literature does not always separate them. The *mode count* is $(p+1)/2$, or 49 at $p = 97$: the number of distinct Fourier frequencies the generalising solution of Gromov (2023) uses. The *order parameter* is the mean inverse participation ratio of the first layer's Fourier spectrum, which is $1.000$ for that solution because each neuron carries a single frequency and about $0.041$ at random initialisation; it measures how concentrated each neuron is, not how many modes exist. The *effective rank* is the participation ratio of the singular values of the first-layer weight matrix, which for the analytic solution is $148.8$: that matrix is $500 \times 194$ and full rank, because each non-zero frequency contributes a sine and a cosine direction in each of the two input blocks, so $148.8$ says how evenly those $194$ directions are used and is not a count of anything. Only the first is a dimension in the sense the closed form fixes. Two things follow from measuring the other two, and both qualify readings of the representation dimension elsewhere.

*The representation is not complete at the transition.* For modular addition the order parameter reads $0.067$ just after memorisation at step $9\,170$, $0.189$ just before generalisation at step $11\,950$, $0.263$ when validation accuracy first reaches $100\,\%$, and then rises for a further $87\,000$ steps to $0.595$ against an analytic $1.000$. It does not jump at the transition; it begins rising at memorisation and passes smoothly through. Generalisation therefore arrives well before the representation is complete, and requires it only to be periodic enough. The effective rank of the same layer carries almost no signal at all: it reads $139.1$ at random initialisation, within seven per cent of the analytic $148.8$ before any training, then spans $132.6$ to $150.9$ over the run without being monotone and overshoots the reference before falling back. Landing near the reference is therefore not evidence of anything.

*The order parameter is only interpretable against its own reference.* Table 16 gives the measured value beside each task's own closed form. Two of the tasks that generalise have a reference near the floor, because the representation is periodic in $g_1(n)$ while the spectrum is taken over the raw index

$n$, and $g_1$ is not linear there. Read against modular addition's 1.000, a measured 0.044 says nothing was learned; read against that task's own 0.062, it says the run has essentially converged. The tasks that never generalise sit at the initialisation floor, and the analysis returns no reference for them rather than substituting another task's.

Table 16: Fourier inverse participation ratio of the first layer at the end of training, against each task's own closed form. All runs are full-batch gradient descent with no weight decay at $p = 97$. A dash marks a task for which the source papers give no closed form; the four tasks below the rule, whose grokking step reads "none", are the ones that never generalise.

| task | measured | own reference | grokking step |
|---|---|---|---|
| $n + m$ | 0.595 | 1.000 | 11 950 |
| $n - m$ | 0.582 | 1.000 | 11 570 |
| $n^2 + m^2$ | 0.093 | 0.052 | 10 380 |
| $n \cdot m$ | 0.073 | — | 11 600 |
| $(4n_1 + n_2^2)^3$ | 0.243 | 1.000 | 12 560 |
| $(2n_1 + 3n_2)^4$ | 0.283 | 1.000 | 7 750 |
| $(5n_1^3 + 2n_2^4)^2$ | 0.044 | 0.062 | 6 430 |
| $(4n_1 + n_2^2)^3 + n_1 n_2$ | 0.040 | — | none |
| $(2n_1 + 3n_2)^4 - n_1^2$ | 0.043 | — | none |
| $(5n_1^3 + 2n_2^4)^2 - n_2$ | 0.039 | — | none |
| $n^3 + nm^2 + m$ | 0.040 | — | none |

## N THE THEILER EXCLUSION OF THE TWENTY-DIRECTION CONFIGURATION

The Theiler exclusion is meant to remove neighbours that are close in time rather than in phase space, and the rule this study asks for is $W_T = \max((E - 1)\tau,\, t_{\mathrm{acf}})$: at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap. The implementation caps $W_T$ at 150 samples, because the neighbour query cost is linear in it. At the eight-direction configuration the embedding span is $(20 - 1) \times 4 = 76$, so the cap binds only where the autocorrelation time exceeds 150: never on the calibration systems, whose autocorrelation time is one to three samples, but on five of the seven parameter-norm records of section 7, where it is 161 to 858. Those five were therefore scored at $W_T = 150$ rather than at the value the rule asks for, in the direction that inflates the estimate. At the twenty-direction configuration the span is $(40 - 1) \times 16 = 624$, so the cap binds unconditionally and the exclusion used is 150, under a quarter of what the rule asks for. Every row of the twenty-direction scores records $W_T = 150$; this is a departure from the stated protocol, not a choice we made deliberately, and it is why section 5.4 marks that result exploratory.

Table 17 measures what it costs, by re-scoring the stored trajectories at both exclusions. Nothing is regenerated, so the systems cannot drift between the arms.

Table 17: The twenty-direction system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured $\mathrm{PR}^{\mathrm{pos}}$ and is not comparable with table 2, which is taken over more observers. What matters is the difference between the columns.

| $r$ | measured $\mathrm{PR}^{\mathrm{pos}}$ | MG at $W_T = 150$ | MG at $W_T = 624$ |
|---|---|---|---|
| 2 | 2.00 | 2.35 | 2.24 |
| 5 | 5.00 | 4.53 | 4.11 |
| 8 | 8.00 | 5.82 | 5.34 |
| 11 | 11.00 | 9.16 | 8.67 |
| 14 | 14.00 | 9.56 | 9.04 |
| 17 | 17.00 | 13.16 | 12.71 |
| 20 | 19.99 | 13.64 | 13.45 |
| median absolute error | | 2.78 | 3.13 |
| Spearman $\rho$ | | +1.00 | +1.00 |

Three things follow. The ordering, which is what section 5.4 claims at this range, is unaffected: the rank correlation is $+1.00$ under both exclusions. The absolute value moves down by $0.11$ to $0.51$ components, so the capped exclusion was mildly inflating it, in the direction that a shortage of exclusion predicts. And the correction makes the recovery slightly worse rather than better, $2.78$ to $3.13$ in median absolute error, so the cap was not flattering the result: enforcing the protocol costs accuracy and buys correctness. The saturation above about six directions is present at both settings and is therefore not an artefact of the exclusion.

The same cap has a second effect that the identifiability ratio inherits. The exclusion actually applied is $W_T = \min\bigl(\max((E-1)\tau,\, t_{\mathrm{acf}}),\, 150\bigr)$. The numerator is computed at $2E_{\max}$ with $\tau$ held fixed, so at the eight-direction configuration its span is $(40-1)\times 4 = 156$ and the denominator's is $76$: the numerator is capped at $150$ whatever the data does, while the denominator is capped only once $t_{\mathrm{acf}}$ reaches $150$. The two arms are therefore treated identically exactly when $t_{\mathrm{acf}} \geq 150$ and differently when it is smaller. On the parameter-norm records of section 7 five of seven have $t_{\mathrm{acf}}$ between $161$ and $858$ and are clean by this test; the asymmetric cases are the two short-correlation records and, more consequentially, the calibration systems of section 5, whose autocorrelation time is one to three samples, so that the reference band of section 6.4 was itself measured with $76$ against $150$. The effect is small at this span and the diagnostic separates the regimes of section 6 by a wide margin, but a ratio built from two differently excluded estimates is not the clean quantity section 3.4 describes, and lifting the cap is the first thing to change in any follow-up.

# O Run inventory

Table 18 lists every training run the paper draws on. Two architectures appear: **T** is the one-layer transformer of Nanda et al. (2023), $d_{\mathrm{model}} = 128$, four heads, no layer normalisation, $226\,816$ parameters on modular addition at $p = 113$ and $228\,608$ on $S_5$, trained in double precision with AdamW at learning rate $10^{-3}$ and $\beta = (0.9, 0.98)$; **P** is the two-layer quadratic perceptron of Gromov (2023), $145\,500$ parameters at $p = 97$ and width $500$, trained by full-batch gradient descent with no regularisation.

Three departures from the source configurations should be stated, since two of them are what put the runs in the regimes section 7.1 assigns them. Nanda et al. (2023) train full batch; we use mini-batches of $256$, which is what makes this setting stochastic rather than deterministic. The $S_5$ weight decay of $0.2$ and the $S_5$ task itself (Power et al., 2022) are outside that configuration. And the two $S_5$ runs step the optimiser twice per batch, reproducing a duplicated update in our own earlier released logs, so their effective learning rate and decay are about double the nominal ones.

**What counts as an event.** Memorisation and generalisation are the first logged step at which the training and the validation accuracy respectively reach $95\,\%$, with no persistence requirement. Training accuracy is measured on the current mini-batch and validation accuracy on a fixed $512$-example subset of the validation split, so both carry a sampling error of a few tenths of a per cent, and the step is resolved only to the logging stride, which is $10$ steps for the modular runs and $5$ for $S_5$. The extended reruns of section 7.2 instead require the threshold to hold over a sustained block, which is the stricter rule; the two put memorisation within $70$ steps of each other on every run here, and differ by $1\,080$ steps on one generalisation step; no claim turns on either. A dash means the criterion was not met inside the budget. Nothing here establishes that such a run would never generalise, and table 18 records the final validation accuracy so that the reader can see how far from the threshold each one ended.

Table 18: Training runs used in this paper. $\alpha$ is the training fraction, "final" the validation accuracy at the end of the budget. A dash in the generalisation column marks a run that had not generalised when its budget ended; such runs are censored observations, not established negatives. The name `s5_wd1` denotes the weight-decay arm of the $S_5$ pair, whose decay is $0.2$; it is not a decay of $1$.

| run | task | arch | wd | $\alpha$ | budget | mem. | gen. | final | used in |
|---|---|---|---|---|---|---|---|---|---|
| *Trajectory sketch attached* | | | | | | | | | |
| `mod_wd1` | mod. add. $p$=113 | T | 1.0 | 0.30 | 20k | 1 470 | 13 700 | 1.00 | section 7.3 |
| `mod_wd1_s43` | mod. add. $p$=113 | T | 1.0 | 0.30 | 20k | 1 580 | 6 660 | 1.00 | section 7.3 |
| `mod_wd1_s44` | mod. add. $p$=113 | T | 1.0 | 0.30 | 20k | 1 410 | 3 680 | 1.00 | section 7.3 |
| `s5_wd1` | $S_5$ composition | T | 0.2 | 0.50 | 15k | 1 280 | 6 735 | 1.00 | section 7.3 |
| `mod_wd0` | mod. add. $p$=113 | T | 0.0 | 0.30 | 20k | 630 | — | 0.03 | section 7.3 |
| `s5_wd0` | $S_5$ composition | T | 0.0 | 0.50 | 15k | 1 015 | — | 0.02 | section 7.3 |
| *Extended reruns,* 120 000 *steps* | | | | | | | | | |
| `grokpos_s0` | mod. add. $p$=113 | T | 1.0 | 0.30 | 120k | 1 450 | 5 110 | 1.00 | section 7.1 |
| `lowdata15_s0` | mod. add. $p$=113 | T | 1.0 | 0.15 | 120k | 620 | 110 940 | 0.99 | section 7.2 |
| `lowdata15_s1` | mod. add. $p$=113 | T | 1.0 | 0.15 | 120k | 630 | — | 0.94 | section 7.2 |
| `lowdata15_s2` | mod. add. $p$=113 | T | 1.0 | 0.15 | 120k | 660 | — | 0.01 | section 7.1 |
| `lowdata20_s0` | mod. add. $p$=113 | T | 1.0 | 0.20 | 120k | 800 | 39 600 | 1.00 | section 7.2 |
| `wd0_s0` | mod. add. $p$=113 | T | 0.0 | 0.30 | 120k | 610 | — | 0.09 | section 7.1 |
| `wd0_s1` | mod. add. $p$=113 | T | 0.0 | 0.30 | 120k | 630 | — | 0.07 | section 7.1 |
| `p211_wd0` | mod. add. $p$=211 | T | 0.0 | 0.16 | 200k | 1 800 | — | 0.04 | section 7.2 |
| *Perceptron,* $p = 97$, *full batch, no regularisation* | | | | | | | | | |
| `a_add` | $n + m$ | P | 0 | 0.5 | 100k | 9 170 | 11 950 | 1.00 | section 7.2 |
| `a_mul` | $n \cdot m$ | P | 0 | 0.5 | 100k | 9 130 | 11 600 | 1.00 | section 7.1 |
| `a_sub` | $n - m$ | P | 0 | 0.5 | 100k | 9 040 | 11 570 | 1.00 | appendix M |
| `a_sq_sum` | $n^2 + m^2$ | P | 0 | 0.5 | 100k | 8 360 | 10 380 | 1.00 | appendix M |
| `a_sum_sq` | $(n + m)^2$ | P | 0 | 0.5 | 100k | 7 280 | 8 150 | 1.00 | appendix M |
| `x_mix_quad` | mixed quadratic | P | 0 | 0.5 | 100k | 9 630 | — | 0.01 | figure 6 |
| `x_no_grok` | $n^3 + nm^2 + m$ | P | 0 | 0.5 | 100k | 9 760 | — | 0.01 | section 7.2 |
| `g_p1` | $(4n_1 + n_2^2)^3$ | P | 0 | 0.5 | 100k | 7 410 | 12 560 | 1.00 | section 7.2 |
| `g_p1x` | $+ n_1 n_2$ | P | 0 | 0.5 | 100k | 10 050 | — | 0.01 | section 7.2 |
| `g_p2` | $(2n_1 + 3n_2)^4$ | P | 0 | 0.5 | 100k | 6 520 | 7 750 | 1.00 | section 7.2 |
| `g_p2x` | $- n_1^2$ | P | 0 | 0.5 | 100k | 9 780 | — | 0.02 | section 7.2 |
| `g_p3` | $(5n_1^3 + 2n_2^4)^2$ | P | 0 | 0.5 | 100k | 5 880 | 6 430 | 1.00 | section 7.2 |
| `g_p3x` | $- n_2$ | P | 0 | 0.5 | 100k | 9 430 | — | 0.73 | section 7.2 |

Four further runs repeat the top four perceptron rows with the trajectory sketch attached, at full batch and again with mini-batches of 512, for the measurement of appendix J; a third pair repeats them at $p = 23$. Their memorisation and generalisation steps agree with the rows above to within one or two logged steps, which is what makes the comparison of appendix J a comparison of noise rather than of learning.

Three notes on how the censored rows should be read. `lowdata15_s1` ends its budget at $94\,\%$ validation accuracy and still rising, so it is the clearest censored observation in the set rather than a negative. The generalisation step of `lowdata15_s0` is 110 940 under the sustained-block criterion used for the extended reruns and 109 860 under the plain first-crossing rule used elsewhere; nothing in the paper depends on the difference. And `g_p3x` reaches $73\,\%$ validation accuracy against a majority-class baseline of $1.3\,\%$ while being provably outside the class the architecture can represent, so it is the one row in this table that neither of the source papers accounts for.